An Adaptive AI Agent for Academic Policy Assistance with Secure Multi-User Memory and Controlled Knowledge Evolution Using Zero-Knowledge RAG

Adish Kumar S	**(22z206)**
Hemanthkumar V	**(22z225)**
Jayavarshini S S	**(22z227)**
Prateekshaa T	**(22z246)**
Praneeth M	**(22z279)**

Faculty Guide: Dr. Saranya K G

19Z820 - PROJECT WORK - II

report submitted in partial fulfillment of the requirement for the award of degree of

BACHELOR OF ENGINEERING

Branch: COMPUTER SCIENCE AND ENGINEERING

Of Anna University

April 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

An Adaptive AI Agent for Academic Policy Assistance with Secure Multi-User Memory and Controlled Knowledge Evolution Using Zero-Knowledge RAG

Bona fide record of work done by

Adish Kumar S	**(22z206)**
Hemanthkumar V	**(22z225)**
Jayavarshini S S	**(22z227)**
Prateekshaa T	**(22z246)**
Praneeth M	**(22z279)**

Dissertation submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: COMPUTER SCIENCE AND ENGINEERING

of Anna University

APRIL 2026

Dr. Saranya K G

Faculty Guide

Associate Professor

Department of Computer Science and Engineering

Dr. Sudha Sadasivam G

Head of the Department & Professor

Department of Computer Science and Engineering

Certified that the candidate was examined in the viva voce examination held on ……………

Internal Examiner

Panel Members

CERTIFICATE

Certified that this report titled **An Adaptive AI Agent for Academic Policy Assistance with Secure Multi-User Memory and Controlled Knowledge Evolution Using Zero-Knowledge RAG** for the **Project Work** (19Z820) is a bonafide work of **Adish Kumar S (22z206), Hemanthkumar V (22z225), Jayavarshini S S (22z227), Prateekshaa T (22z246), Praneeth M (22z279)** who have carried out the work for the partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering.

Place: Coimbatore
Date:

Members	**Faculty guide**
Adish Kumar S	Dr.Saranya K G
Hemanthkumar V	Associate Professor
Jayavarshini S S	Department of Computer Science and Engineering
Prateekshaa T	PSG College of Technology
Praneeth M	Coimbatore - 641004

COUNTERSIGNED

Dr. Sudha Sadasivam G
HEAD
Department of Computer Science and Engineering
PSG College of Technology, Coimbatore

Table of Contents

Acknowledgement

We wish to express our heartfelt gratitude to our esteemed **Principal, Dr. K. Prakasan**, for providing us with the opportunity and essential facilities to successfully complete this dissertation work.

We are immensely grateful to **Dr. Sudha Sadasivam G, Head of the Department, Professor**, Computer Science and Engineering, for her unwavering support, insightful guidance, and constant encouragement throughout this journey.

We would like to thank our **Faculty Guide, Dr. Saranya K G, Associate Professor**, who offered invaluable assistance and motivation throughout the course of this project, allowing us to be engaged in academics from the very beginning.

In addition, our thanks go to the **Program Coordinator, Dr. Arul Anand N, Professor**, who provided valuable help and guidance in the preparation of this dissertation. Furthermore, our **Tutor, Dr. Anisha C D, Professor**, is worth mentioning for her motivational assistance and useful suggestions, which have been helpful in the preparation of this dissertation.

Finally, we take the opportunity to thank all the other students and faculty members of the Computer Science Department for their support in completing this dissertation.

Synopsis

The project is aimed at the creation of a Zero-Knowledge Adaptive AI Agent that will assist students and faculty in getting answers to academic policy inquiries. The point is that it should develop a system that could provide clear and trustworthy answers relying on institutional policy documents, without referring to general model knowledge. It is meant to operate in a secure manner, it can support several users and also adapt when policies are modified with time.

The system is developed using FastAPI and React. It stores policy content in smaller embedded components using ChromaDB, and user data, sessions, messages, preferences, memory details, extensions, and records of policy updates using MongoDB. When a person raises a question, the system will search the most related policy data, listen to the context of the conversation and produce an answer based solely on the content that has been retrieved.

Some adaptive features are also included. Users are required to undergo authentication and identity checks. The system is able to modify the responses according to the preference of the user. It also lets users manage their personal data, thus they can view, retain or delete information stored. Administrators and faculty receive access to additional tools that aid in certain academic activities.

The other noteworthy aspect of the project is the way of managing policy changes. When a user declares a policy outdated, the system can create a flag and gather evidence, and generate a review request. It then determines which sections of the policy stored are impacted. Approved reviewers can intervene, revise the material, and maintain a history of the changes to be transparent.

Altogether, this project unites safe information recall, interaction being user conscious, easy access to personalization, and a regulated means to refresh knowledge. The outcome is a system that is capable of supporting academic institutions in a reliable and scalable manner.

CHAPTER 1

INTRODUCTION

.

There are numerous rules, processes and other policy documents that run academic institutions. These include exams, in-house tests, course enrollment, revaluation, internship, project and student progress in general. Although all this information is officially documented, it is not always easy to find what is required by the students or faculty. Some find it hard to find the appropriate rule or even get the entire steps involved, particularly when they require the information fast.

In practice, policy information is dispersed in various locations such as portals, notices, circulars and department messages. Due to this, it may be slow and confusing to find correct information. Even those questions, which could be answered directly with references to existing documents, people tend to ask the staff or make their own interpretation.

With the rise of Large Language Models, conversational systems have improved a lot. They are able to comprehend natural language questions and answer them in a straight forward manner. Nevertheless, these models themselves cannot be completely used in academic policies. They can provide responses which are not justified or even false when not connected to trustworthy sources. Retrieval Augmented Generation comes to the rescue here. It links the model to real documents, therefore, responses are made on the basis of what the model actually remembers rather than merely what is in the model. In academic career, this is very important since the responses should be true, traceable, and restricted to official data.

As well, the current academic support tools are supposedly supposed to do more than serving as simple answers to frequently asked questions. They must deal with logged-in users, comprehend various roles, maintain conversations, and provide a certain degree of customization. Meanwhile, they also need to handle user information with caution and enhance knowledge in a regulated manner when policies evolve. Research indicates that user-adaptive systems can be more helpful and this can be achieved only when privacy, safety and accuracy of facts are preserved. This project is based upon that requirement, to construct a more dependable and managed academic assistant to be used in real institutions.

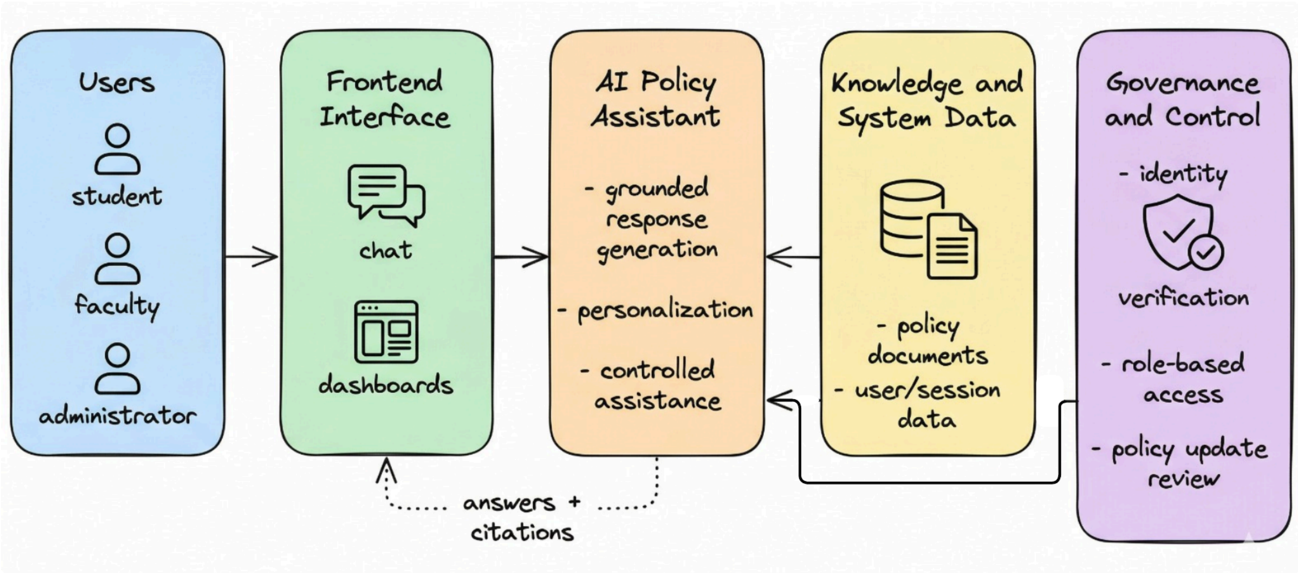

Fig. 1.1: Conceptual big picture of the zero-knowledge adaptive academic policy assistant

The proposed system in this project is a Zero Knowledge Adaptive AI Agent that is proposed to assist in academic policy queries. Simply put, it combines a retrieval-based method, in which the responses are extracted out of real policy documents, with an AI model, which can give a clear explanation of the responses.

This is in addition to the fact that the system has secure login and identity checks; hence only people who are authorized can access the system. It also aids user preferences and memory, which is useful in providing more pertinent responses with time. It has other extension features that enable it to do certain academic tasks when necessary.

The other significant aspect is its policy update management. Rather than it just automatically changing information, it is a human-in-the-loop process, with any updates reviewed and approved before implementation. Fig. 1.1 gives a rough description of the functionality of the system and Table 1.1 links the overall project objectives to the various modules employed to accomplish them.

1.1 Background and Motivation

The development of online systems in institutions of learning has exposed much information to the internet. However, this does not necessarily imply that it can be easily accessed or used. It is common to see students go through a number of documents before they could answer simple questions. As an illustration, they might wish to find out whether Continuous Assessment marks are revaluable, how exam registration is done or what happens with add, drop or withdrawal provisions. As long as the information is available, it still takes time to locate and comprehend it particularly when the person being assimilated is not conversant with the working principles of such processes. This has resulted in individuals taking a lot of time than necessary and not all individuals interpret the rules in the same manner.

This can be facilitated by conversational AI that makes the process more straightforward. Users do not have to scan files to find answers; all they have to do is to ask questions. However, it can only be effective when the system remains true in its sphere. Any little error can lead to misunderstanding of deadlines, eligibility, or necessary actions in the case of academic policy use. The system must therefore not only be smart, it must be careful and controlled. This is the reason why zero knowledge retrieval approach is employed. The system responds to documents that are verified and in cases where it fails to get appropriate support, it does not provide an approximation.

Meanwhile, there is an increased expectation of such systems by the users. They desire it to know their likes, keep the conversation flowing, and modify responses according to their circumstances. It may be that additional tools are required of faculty and administrators to complete certain academic tasks and not merely basic answers. And on top of that the policies do not remain the same. The institutions revise and issue new circulars and revisions of rules. All this makes it obvious that not just a chatbot is sufficient. What is required instead is a more adaptable and regulated academic support system that is capable of adapting and at the same time remaining dependable.

1.2 Problem Statement

Institutions of learning tend to publish policy information as hardcopy documents, notifications or portal logs. These are formal and dependable, yet they are not designed to be easily interacted with. Citizens cannot simply pose questions, in a natural manner, or receive prompt clarifications. Therefore, whenever students or faculty questions exams, grading, revaluation or project regulations, it usually requires them to go through documents independently. This may be a long process and it is not difficult to misinterpret something.

Not using a complex chatbot is not the entire solution to this problem. When the chatbot is relying on a general language model, it may provide answers that may appear correct but which are not really grounded on official policies. Conversely, when it can only extract information in documents without any conversational skills, it might not respond to follow up questions or respond to the context. It will not be personalized either. Most of the systems that are in place do not attach much attention to such important features as secure login, user specific data handling or a proper update of the policies as the changes occur.

Due to this reason, the project is aimed at developing a system not only intelligent, but also reliable and controlled. The objective is to develop an academic policy assistant which responds exclusively to approved institutional sources, provides responses to multiple authenticated users, and provides responses which is sensitive to context and user preferences without losing accuracy. It must also be capable of managing policy changes in a systematic manner, particularly when users report changes or potential inconsistencies.

1.3 Objectives of the Project

The primary objective of the project is to develop an academic policy assistant based on AI that operates in a safe and regulated manner. It is based on a zero knowledge retrieval model, i.e. it responds to questions based on verified policy documents. Meanwhile, it helps accommodate several users and enables the system to update its knowledge in a cautious and monitored fashion. It is not merely to provide answers to questions, but to create a whole, dependable academic support system.

To do this, some important goals are outlined. A significant measure is constructing a retrieval-based system on top of policy documents with ChromaDB with FastAPI as the server. The system is set up to strictly adhere to zero knowledge behavior and therefore will respond only through retrieved evidence and not respond when adequate support is lacking.

Access to users is also done with caution. It has built-in secure logging by using Google OAuth and role-based controls on the server. It also allows personalization, whereby user preferences and memory are handled in a secure manner among users. Non-admin users undergo an AI-based identity check step before they can access it regularly.

The system provides faculty and administrators with additional tools in addition to basic question answering. These tools might be implemented via prompts or scripts, depending on the activity. The other important characteristic is the process of updating the policies. The system enables reviewing and approval of changes by humans before any information is updated when changes are reported.

Lastly, appropriate validation is also a concern in the project. The policy data and the system behavior is both tested through structured means, as well as with the aid of language models. The link between these objectives and the system modules is presented in Table 1.1.

Objective	Related Module / Feature	Expected Outcome
Zero-knowledge academic question answering	RAG pipeline with ChromaDB retrieval and grounded generation	Accurate answers based only on institutional policy evidence
Secure authenticated access	Google OAuth, JWT authentication, and role-aware routing	Controlled access to the platform for different user categories
Adaptive personalization	Preference extraction, preference application, and user memory handling	More relevant and user-aware responses without losing factual grounding
Identity verification	AI-assisted identity verification workflow	Trusted access control for non-admin users before normal usage
Extension-based academic support	Prompt-based and script-based extension framework	Domain-specific tools for faculty and administrators
Controlled knowledge evolution	Policy claim detection, proof submission, ticket review, and chunk deprecation workflow	Human-reviewed updating of outdated policy knowledge
System validation and quality assurance	Dataset validation, functional testing, and workflow evaluation	Improved reliability and credibility of the academic support system

Table 1.1: Mapping project objectives and the significant system modules that will be implemented in the proposed platform.

1.4 Scope of the Project

This project is focused only on academic policy support within a controlled institutional setup. The system employs chosen policy documents that are kept in markdown files. The documents are processed and converted into embeddings and stored in a vector database in order to be accessed when a user poses a query. It primarily deals with exams, grading, course registration, revaluation, and projects, internships and such like academic procedures.

Several parts are covered in the work dealt with in this project. It processes documents (policy files are divided into smaller parts and embedded and stored to be searched). Upon receiving a query, the system will find the corresponding content and produce responses on the basis of that, with reference sources, a sense of confidence, and the capability to reject unsupported queries.

It also allows a secure log in and also keeps track of user sessions. Conversations in the past can be recalled by the system and this will assist in the follow up questions being handled more naturally. Simultaneously, it enables a certain degree of customization, via user preferences, and yet, keeps memory utilization managed and safe.

Faculty and administrators can have additional features at extensions that assist with particular academic activities. Policy update process is another significant component. Users are able to provide evidence when they believe that something has changed and this undergoes a review process. The system detects content that is affected and it updates in a controlled manner keeping the appropriate records of all updates.

Nevertheless, the project has definite boundaries. It is not intended to be a general purpose educational assistant. It does not usurp official academic authorities and it does not revise policies automatically without human intervention. The broader capabilities such as access to open web knowledge or predicting academic risks are not in the present scope and could be an option to be taken into consideration in the further work.

1.5 Proposed Solution Overview

The suggested system is a Zero Knowledge Adaptive AI Agent which assists in responding to questions related to academic policy in a regulated and trustworthy manner. It operates off a retrieval-based strategy, in which it initially identifies the most pertinent sections of policy documents in ChromaDB. It then combines them with such other things as the context of prior conversations and, where possible, preferences of the user to create a response. The most important thing is that it is this information that is retrieved that generates the answer. When the system is not able to locate sufficient supporting evidence, it just refuses to provide an answer rather than make a guess or fabricate information. This would minimize the number of errors and make the responses reliable.

The solution is based on a fundamental retrieval augmented model but incorporates a number of helpful features. Firstly, it offers support to several users with the help of secure logins and role-based accessibility. It is also capable of modifying responses depending on user preference and stored memory, however this is very cautiously done so as to prevent abuse.

The other significant characteristic is identity verification. This step is obligatory to non-admin users prior to granting them full access, providing an additional trust layer. Faculty and administrators also can take advantage of the system through special extensions that are capable of performing special academic tasks not in question answering.

Lastly, the system has an organized approach to policy changes. In case a user identifies a potential change or inconsistency, the system may identify it and gather evidence and forward it to be reviewed. The claim can then be verified by authorized users and the knowledge base updated in a controlled manner and records of all changes maintained.

1.6 Organization of the Report

The remainder of this report is organized in a straightforward order. Chapter 2 examines related work, including such areas as retrieval augmented generation, conversational academic systems, and safe management of user memory, identity verification, human-in-the-loop updates, and verification with language models.

Chapter 3 shifts to the side of the system design. It describes the requirements, various user roles, the overall architecture, technologies involved, and the practicality of whether to implement the system or not. Subsequently, Chapter 4 analyzes the existing solutions and identifies their shortcomings and this justification as to why this new approach was necessary.

Chapter 5 dwells upon the proposed system in more detail. It describes the processing of documents, retrieval in zero knowledge state, and the generating of answers in a grounded manner. It also addresses attributes such as personalization, authentication, extensions, and management of updates to policies in a controlled way.

Chapter 6 then explains the real construction of the system, which entails the frontend, backend and the data storage components of the system. Chapter 7 considers testing and validation, including checking the dataset, testing system functions, examining retrieval performance, and checking workflows.

Lastly, Chapter 8 talks about the results and what the system can accomplish, and Chapter 9 concludes the report and recommends potential future improvements.

CHAPTER 2

LITERATURE SURVEY

2.1 Retrieval-Augmented Generation in Domain-Specific Question Answering

RAG, or Retrieval Augmented Generation, has become a beneficial method to make AI responses more trustworthy. The system does not solely rely on what the model already knows, but first attempts to retrieve relevant information by accessing an external source. That content retrieved is then assembled into the final answer. This simplifies the process of finding the source of the answer and minimizes the possibility of providing unfounded and wrong answers, especially in situations where the correct information is of utmost importance.

This style is quite suitable in such spheres as academic policy support. The system does not require a good general knowledge as most rules and procedures are already in document form. It only should find and employ the appropriate policy content. Attention to these sources helps the assistant not to exceed its limits and go astray. It also assists to diminish the issue of hallucination whereby the system would otherwise create wrong information in a dialogue.

Simultaneously, a basic RAG setup is not sufficient to a complete academic support system. The majority of simple variants merely involve retrieving data and produce responses. When changes occur, they typically do not deal with issues such as secure user access, personalization or revising policies. Due to this, the project is based on RAG, though, it incorporates additional functionality aiming at user awareness and adequate control over the development of the system.

2.2 Conversational Academic Support Systems

The concept of conversational systems in education and institutions is now common to assist students in answering frequent questions, administrative questions, and general advice. The first benefit is that users can just ask questions in natural language without searching the portals or reading long documents. This can make academic policy-related issues such as exam practices, grading policies, rework policies, or project policies, far more understandable in the case of academic policies.

Nevertheless, a lot of these systems are limited. Others are constructed in the form of simple FAQ tools, where the responses are based on keywords. These tend to fail when the user poses some follow up questions or refers to the previous context. Conversely, other systems rely greatly on general language models. They may be fluent, but may provide answers that are not in fact backed by official information.

Studies indicate that conversational systems are more effective when the responses are linked to appropriate background sources, rather than simply creating answers on their own. It is particularly critical within the academic context, where precision is a factor. Due to this, an effective academic assistant must be flexible in the dialogue and well-documented. One of the major concepts of the system presented in this project is bringing these two together.

2.3 Personalization and Secure Multi-User Memory

Dialogue system personalization is aimed at making response more relevant to an individual user. This may involve the user, his or her preferences, the way they tend to communicate and what they have requested previously. Research indicates that in systems where this type of user-specific context is employed, dialogues can feel more comfortable and productive.

Meanwhile, introducing memory to AI systems is associated with certain difficulties. Storing user data may be useful in ensuring continuity in interactions, yet it is also problematic. It has privacy risks, risks of confusing the data between users or risks of storing excess information without valid control. Studies indicate that although memory has the potential of enhancing a long-term understandability, it must be managed well to prevent issues such as data leakage or unwanted accumulation of information.

These concerns are even more crucial in an institutional context. Users can interact with the same system in many instances with different roles and access levels. Due to this, memory cannot be approached in an open or uncontrolled manner. This project is much more cautious as it provides user-specific memory and preferences, but the control is evident at the user level. Rather than considering memory as being hidden within the model, it is handled transparently and in a controlled fashion.

2.4 Identity Verification and Access Control

Just a mere system of simple login in academic institutions is not usually sufficient. The system must also be certain of the identity of who the user is, and what category they fall into, like the student, faculty or administrator in most instances. This is crucial in cases where users are granted access to such things as policy updates, administrative tools or services that require their institutional identity.

The general conversational AI systems are primarily concerned with easiness of interaction. Security and adequate access control are of equal importance however, in an academic environment. Due to this fact, authentication and identity verification are not considered as optional features, but rather as vital components of the system. This is among the main distinctions among a controlled academic assistant and a normal public chatbot.

In a wider sense, user data are quite sensitive and should be handled with care. Although this project does not apply more sophisticated techniques such as differential privacy directly, it is still based on the concept that the information of the users must be secured and utilized in a restricted manner. Surveys conducted in this field also confirm this perception by establishing that systems must handle identity, access, and data exposure in a mindful and conscientious way.

2.5 Human-in-the-Loop Knowledge Updating

A typical issue with the static knowledge systems is that the information may become obsolete. Policies in academics are not static. New circulars can be issued, time limits can vary, and rules or instructions regarding evaluation or instructions to departments can be revised. Once this occurs, previous policy content might cease to be right. Although a chatbot may be founded on trusted documents, they may still provide outdated information when the knowledge base is not regularly updated.

This ties into a bigger concept in AI where systems require a means of eliminating or updating old knowledge without disrupting the functionality of all the rest. However, this process may be risky when it is done automatically, particularly in academic settings where the accuracy is valued and the claims of users must also be verified. Due to this, a fully automated update process is not necessarily the most appropriate option.

A more secure method is to resort to human inspection. It is also proposed in research that systems employed in sensitive or controlled places should not update themselves without adequate control. That is being done in this project by a systematic policy update procedure. In case an individual reports about a potential change, the system gathers evidence, verifies what aspects of the policy are involved, and forwards it to review. The only authorized users can approve the updates, remove the old content, and change. Every update is followed and, therefore, there is a distinct history of changed and the reasons.

2.6 LLM-Based Validation and Evaluation

Answers are no longer just generated using large language models. Data and system outputs are also being checked, reviewed and evaluated using them. This type of validation comes in handy in this project in two aspects. To begin with, it assists in verifying the academic policy dataset as such, ensuring that it is consistent, complete and can be used within a retrieval-based system. Second, it aids in verifying the behavior of the application, including whether answers are grounded, clear, and adhere to the proper procedures.

This is significant since institutional data is usually ready or modified prior to its application in the system. An additional check can be offered by using an LLM to check whether the documents are organized well and can be retrieved.

Meanwhile, such validation cannot be considered entirely reliable itself. It is an aid and not an ultimate power. The system, in its entirety, remains as correct as the retrieval process is designed, the quality of the original documents, and the constraints imposed on the manner in which the model produces the responses.

2.7 Research Gap and Motivation for the Proposed System

The literature demonstrates that there are numerous helpful ideas already in this sphere. By way of example, retrieval augmented generation assists in the generation of responses on the basis of actual files, personal dialogue is enhanced by user specific dialogue, and memory based systems aid in continuity. The methods also exist to update knowledge in case information varies.

Nonetheless, they tend to be studied and constructed individually. There are systems that are primarily oriented on searching out the right information but are unable to customize or individual requirements. Others make attempts to individualize answers, but fail to tie answers to credible sources. Similarly, most conversational systems lack an appropriate mechanism of updating knowledge in the event of changes in policies or the emergence of contradictions.

Due to this, an evident gap is evident in terms of academic policy support. The efficient system in this environment cannot be based on a single or two characteristics. It must unite a number of things simultaneously. This involves accessing data in official records, making conversation flow conveniently over time, to multiple users, and to be able to personalize with restraint and appropriate memory management. It must also check identity of users, assist additional academic work tools and a built in mechanism of updating knowledge that is man-reviewed.

This project attempts to unify all these aspects into a single system rather than dealing with them individually. The comparison between these ideas from the literature and the proposed system is shown in Table 2.1.

Literature Area	**Key Strength**	**Limitation in Isolation**	**Relevance to Proposed System**
Retrieval-Augmented Generation	Grounded answer generation from external documents	Does not by itself provide user-aware security or governance	Forms the core zero-knowledge policy-answering pipeline
Conversational academic assistants	Natural language access to institutional support	Often weak in factual grounding or multi-turn control	Motivates user-friendly academic interaction
Personalized dialogue and memory systems	Adaptive responses and conversational continuity	Can introduce privacy and control risks if unmanaged	Supports controlled preference and memory framework
Identity verification and access control	Trusted and role-aware usage	Not usually integrated with RAG academic assistants	Strengthens secure institutional deployment
Human-in-the-loop knowledge updating	Safer correction of evolving knowledge	Adds workflow complexity and review dependence	Enables controlled policy evolution in the project
LLM-based validation	Useful quality assessment for datasets and outputs	Cannot replace authoritative human judgment	Supports dataset and application evaluation

Table 2.1: Comparative overview of key literature focus and its applicability to the proposed academic policy assistant.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Functional Requirements

The functional requirements of this system explain what the system should be able to do to facilitate academic policy support. The system will not just answer some basic questions; since the aim is to create a controlled platform to be used by the institution, the system will do more than just that.

On a simple level, the system is supposed to enable the authenticated users to pose questions using natural language and have the answers according to official policy documents. In order to do this, it must be able to retrieve relevant content, produce responses based on that content, display the source and not to answer when the evidence is not available. It must also be able to deal with multi-step conversations, such that follow up questions may be interpreted relative to previous questions.

In addition to the general chat feature, the system has user-specific features. It logs the history of the sessions and lets the users control their preferences. It also manages memory in a controlled manner, users are able to see, store or delete stored personal details. To make it even more secure, other users cannot access some features unless they are verified as an administrative user.

On the administration side, the system upholds role-based access. Extra tools are available to faculty and administrators to support specific academic activities, either by using prompts or scripts. There is also a policy issue reporting process as part of the system. Users may provide evidence in case a contradiction was discovered, providing a ticket of review. The affected aspects of the policy are then recognized by the system and updates are not performed until humans review them.

Concisely, it is anticipated that the system will respond to academic policy queries in a zero-knowledge manner, maintain a conversation, supply concise sources and trustworthiness, decline unsubstantiated queries, assist in secure access and checks on identity, permit controlled personalization, facilitate advanced task extensions, and address policy changes in a regulated and monitored fashion.

3.2 Non-Functional Requirements

Non-functional requirements are the requirements that specify the fitness of the system to do something. This project is targeted towards controlled academic usage and as such, reliability and safety are more of importance than open-ended response generation.

One of the most important aspects is accuracy. The system must not make unfounded claims and only respond to the content of the academic policy that has been retrieved. Traceability is also significant along with this. The users must be capable of seeing where the answer is originating and in this way they will be able to trust the information.

Security is another key concern. The system archives the user sessions, preferences, memory information, as well as the workflow (admin level). Due to this, it needs to have proper login, separation of roles and good data isolation among users. Admin features ought to be inaccessible to regular users, and personal data should be secured.

The aspect of usability is also a factor. The interface must be easy to use and simple to the students, faculty, and administrators. Different sections such as chat, memory, preferences, extensions, and policy updates should be easily navigated by the users.

Other significant qualities are a few. It must be a system that can be maintained, and additional new policies, tools and features can be added without rebuilding the whole. It is also supposed to be scalable to project level i.e. the design can be expanded later in case it is necessary. Responses must be timely to enable fluid interaction. Lastly, the system must facilitate good governance whereby any policy changes will be monitored and can be accessed, rather than being altered without prior notice.

3.3 User Roles and Access Levels

The system is also constructed as a role based academic platform in which various users are granted access into the system in varying degrees. This is significant since the system is not merely a question answering system. It also contains such features as administration, extensions, and the policy update workflows, which must not be opened to all.

There are three main user roles in the system: students, faculty, and administrators. The primary users are students. They get to ask questions to the academic policy through the system, check on their preferences and access their stored memory. They can also make a claim and give evidence of the same in case they observe a potential change in a policy.

Access is higher among faculty members than students. They can access special extensions specific to academic work, in addition to normal policy enquiries. They can also participate in some of the review processes or support activities depending on their authorizations. This position is significant as there are still decisions that require academic judgment and cannot be fully automated.

Administrators have the highest level of control. They handle users, audit system settings, view the content of stored policy, manage extensions and monitor the policy update process. They can also access dashboards and tools that are required to monitor and manage the platform.

All in all, role separation will aid in making the system secure. It makes sure that users can only access what they are permitted to, ensures that user data is well segregated, restricts access to sensitive tools, and helps in safely managing updates to policies.

3.4 System Architecture Overview

The system is developed based on a modular client server architecture and separation of frontend, backend and storage layers are clear. This arrangement aids in maintaining a system logic, user interaction, and data storage in a well-structured and manageable manner.

React and Vite are used to develop the frontend. It offers various interfaces according to the user roles such as chat, preferences, memory views, extensions, policy updates, and administration pages. Most of the core operations are processed using the FastAPI on the backend. This will cover user authentication, session management, retrieving documents, generating responses, creating preferences, extension management, and policy updates.

To store data, the system is based on two key components. ChromaDB is utilized in storing and retrieving academic policy content in a vector format. Instead, MongoDB supports common application information like user profiles, user sessions, user messages, user preferences, user stored memory, user extension information and user policy update requests. Having these two kinds of data separate would make the system cleaner and easier to maintain.

A user query passes through a number of steps when the system is running. It begins by authenticating and loading the session context. Then, user preferences and memory can be applied. The system then looks up pertinent policy information and produces a grounded answer. In case the query identifies a potential problem in policy, the process may switch to identifying claims, reviewing evidence, and developing a review request. The detailed description of this process is presented in Fig. 3.1.

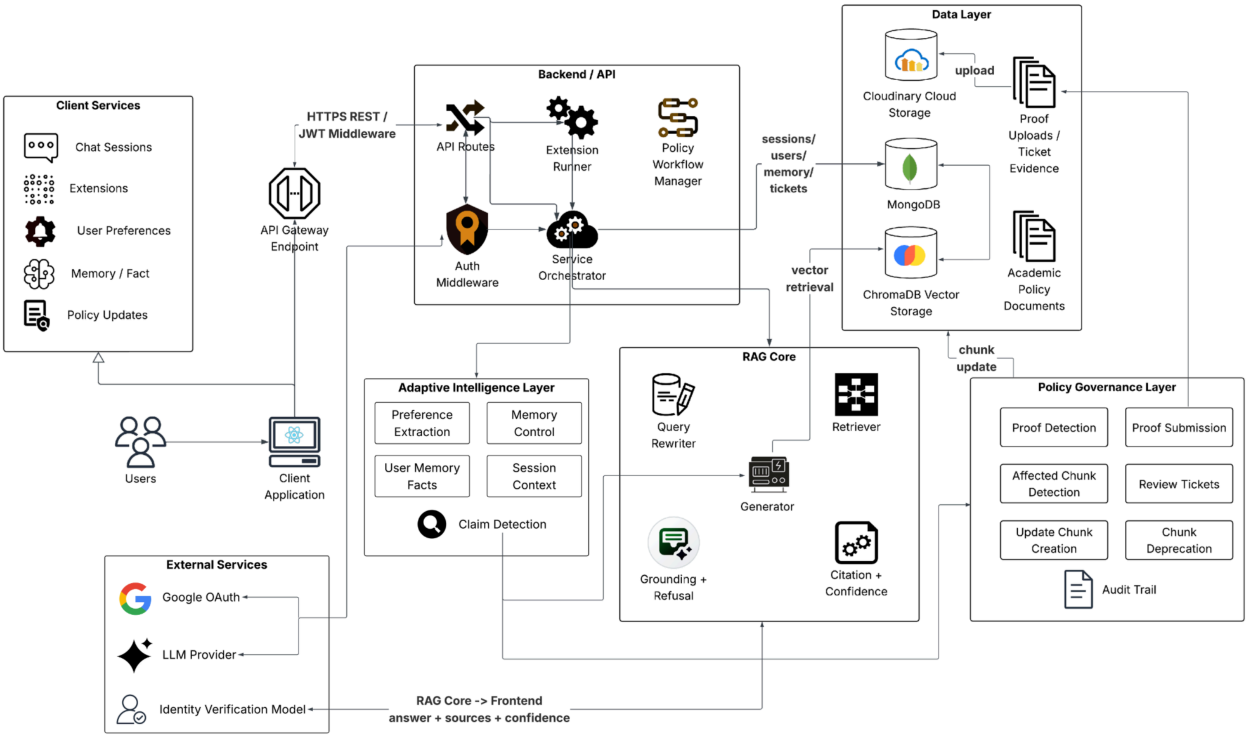

Fig. 3.1: General system layout of the proposed zero-knowledge academic policy support system.

3.5 Technology Stack

The technology stack of the given project was selected to enable modular design, document based search and a web based academic support interface. In general, tools employed work well with a complete stack configuration and align with the objectives of the system.

On the backend side, Python with FastAPI is used. This integration simplifies the construction of APIs and linking AI-related capabilities in a clean manner. As the frontend, we will rely on the React and Vite that assist in the development of responsive pages and splitting the interface into reusable elements.

ChromaDB is utilized as the vector database to deal with knowledge retrieval. The academic policy documents are handled by dividing them into smaller pieces, turning them into embeddings and storing them to search them using similarities. In tandem with this, MongoDB is employed to handle application data. It can be utilized effectively to store such items as user information, user sessions, chat messages, preferences, data in memory and records of policy updates.

The system also has some other technologies. Google OAuth is the one that is used to log in and JWT is the one that is used to manage secure access. The responses are generated through the LLM integration and some validation/extraction activities. There is also a system of custom extension components which assist with faculty or admin related tools. At the frontend, there exist dashboard features that enable users to review policies, review stored memory and manage preferences.

This stack, combined with the AI functionality, and the general platform requirements, is supported. It assists the system to remain safe, flexible and as easy to increase as necessary.

3.6 Feasibility and Constraints

This project can be viewed in terms of technical, operational and practical feasibility. Technically, the system is rather possible as the tools that are used are already existing and popular. FastAPI, React, MongoDB, ChromaDB, and integration of LLM technologies are innovative technologies that can be used to develop a modular academic support system. Moreover, the data processing is also easier with markdown-based policy documents and the retrieval is more manageable.

On an operational level, the system is operational well in an institutional level, particularly in a small or restricted environment. It concentrates solely on academic policy which assists in preventing the risks that tend to accompany open ended AI systems. Such characteristics as role based access and identity verification make it more appropriate to the real academic setting.

Meanwhile, it has some limitations. The quality of responses depends a lot on how complete and accurate the policy documents are. Personalization and memory functionalities should also be handled with care in order to ensure they do not influence the accuracy of policy based responses. Neither can updates to policy be automated, as they need to be properly reviewed in order to ensure that they remain accurate. Moreover, the system is not intended to be deployed on a large scale, rather demonstration and controlled use.

Despite such limitations, the project remains viable to the intended purpose. Its overall design is modular, thus can be enhanced and expanded in the future. Meanwhile, it remains grounded in the real academic needs and is straightforward and clear.

CHAPTER 4

EXISTING SYSTEMS AND THEIR DISADVANTAGES

4.1 Traditional Policy Access Systems

Academic policies are disseminated in most institutional forms such as handbooks, notices, circulars or updates on portals. These are formal and trustworthy yet not necessarily convenient to use when one simply wants to ask a quick question. Typically, a student must read a number of documents, locate the appropriate section and attempt to comprehend it independently. This might be time consuming, particularly when they are not accustomed to the official language or the format of these documents.

Another issue is that these systems don't really "help" actively. They simply sit on the information and it is the user who has to locate what he/she needs. It is impossible to ask any follow-up questions, receive a simple explanation, and clarify any doubts in clear words. Therefore even such simple processes as registration procedures or revaluation policies tend to be negotiated with faculty, office personnel, or even friends.

As such, although these are still valuable documents in the sense that they are official, they do not match the expectations of people today. There's a gap here. The information is available, yet it is not that easy to access and practically use it.

4.2 Generic Chatbots for Academic Support

In educational settings, generic chatbots have become rather widespread. They assist the students to obtain immediate replies, decreasing the workload of staff and addressing simple questions without any considerable time loss. They are okay in simple questions, particularly, the ones that keep reoccurring. It is also user friendly as one can simply type a question and not go through portals or volumes of documents.

Nevertheless, they do not always sit well in the area of academic policies. One issue is that a lot of these chatbots do not even have an official institutional connection. Due to that, their responses can be general knowledge rather than specific policy guidelines. The other problem is that they do not tend to address the user roles, identity checks and customized context that well.

Thus despite making things easier, they may not be able to perform in the realms where precision and control are essential and access is of high quality.

4.3 Conventional RAG-Based Assistants

Retrieval augmented generation systems move a notch further than ordinary chatbots in that they will actually retrieve information within the documents and then respond. They do not rely solely on what the model knows, but rather look at an outside source and utilize that as a way of responding. This renders them more effective particularly in situations where truthful information is of paramount importance.

Nevertheless, despite this enhancement, the majority of these systems remain rather limited. They are primarily created only to retrieve data and come up with solutions. There are several things such as checking who the user is, multiple-user safety, user-specific information and update management over time that are not provided. Due to that, they can be used well in simple use cases or in fixed knowledge bases, but not in the context of a particular environment, such as academic institutions where roles, permissions, and evolving rules matter.

There is one more problem. Such systems typically presuppose that the information that they operate on does not vary significantly. Keep in mind that policies are sometimes updated. When such occurs, most systems lack an appropriate mechanism to deal with it. They might not facilitate conflict checking, gathering evidence, examining updates, and substituting obsolete information in a managed manner. Hence, their usefulness may decline over time, particularly in those spheres where regulations continue to change.

4.4 Limitations of Existing Systems

Based on all this, there are some obvious problems that are evident in the present systems of academic policy.

- Firstly, the conventional document-based systems are not user friendly. Individuals are forced to go through files by themselves and work out things, which may be cumbersome and may be confusing at times. It is not that useful when a person is in urgent need of a solution.
- Next are generic chatbots. They are supportive and provide seamless responses, yet these responses are not necessarily grounded in real institutional policies. So even though the answer may seem correct, it might not necessarily be trustworthy.

- The other gap is in the area of security and user handling. Most systems do not actually authenticate users or modify responses according to roles. Such things as identity verification or role-based access are usually absent, which is an issue in structured environments.
- It also does not address personalization in a good way. Other systems attempt to customize, yet they do not provide users with adequate control of their data or preferences. It turns out to be either too narrow or not very transparent.
- There is also the problem of update of policy. Information is considered fixed by most systems. In case of a change, there is no outlined procedure that can be used to report, review as well as update the system accordingly. That renders it difficult to maintain things in the long run.
- And lastly, such systems are often very inflexible. They do not permit the addition of specialized tools or features that may be required by faculty or administrators, so they are not as useful.

In general, the presence of a chatbot or even a number of documents is not sufficient. The only thing required is a system capable of drawing information based on trusted sources, supporting various users safely, scaling up more, and maintaining updates in a controlled manner as well.

4.5 Need for the Proposed System

The suggested system will address the issues observed in the current methods, but will retain the good parts. Official documents will always be important as academic policies. However, simultaneously, individuals ought to have an opportunity to obtain that information with a little less effort, such as posing a question in simple terms.

This is assisted by conversational AI but it must not answer at will. The answers must remain linked with real policy content. Similarly, personalization may make the system more useful, yet, it must be approached cautiously to ensure that user information remains secure and within control. Policies are dynamic as well, and updates cannot occur automatically without checks. Proper review and tracking is required.

This system is therefore a combination of various useful components in a single space as opposed to having to use only one idea. It employs document-based retrieval to ensure answers remain grounded. It has access controls such as login and identity checks. It also aids user preferences and memory in a secure manner, thus responses become more relevant. Additionally, it lets one add special tools to do academic work, and it manages policy changes with an appropriate review procedure.

Due to this combination, the system is more suited in an academic environment. It is more controlled, reliable and flexible, and at the same time more than just a portal or a simple chatbot.

CHAPTER 5

PROPOSED SYSTEM AND METHODOLOGIES

5.1 Overall Proposed System

It is a simple Zero Knowledge Adaptive AI Agent that is proposed to assist in academic policies within an institution. It is not only intended to answer questions. It also attempts to unite all that under a single roof such as obtaining secure user access, personalized interaction, special academic tools and a decent method of managing policy updates.

On the simplest level, the system consists of a limited number of parts. It has a React user interface, a FastAPI server to process the logic, ChromaDB to store and retrieve policy content, and MongoDB to store user data and system activity. A search system will look through the relevant sections of the policy when the individual poses a question, it will engage the information and then provide an answer as it appears with regards to the context, with no extra information being added. When it fails to get sufficient supportive evidence, it merely fails to answer rather than make an educated guess. This makes the answers secure and dependable, a factor that is crucial in the academic context.

On top of this, there are additional features added to the system making it more practical. It facilitates the process of logging in and identity verification, tracks user preferences and manages memory in a disciplined manner. It also enables the addition of special tools in academic work and provides a process of updating of the policies in case of necessity through review.

In general, it is not a chatbot. It is closer to a system of academic support that is highly organization and is not only based on documents but is also flexible enough to accommodate the actual user requirements. Fig. 5.1 illustrates the flow of the system in general.

5.2 Zero-Knowledge RAG Pipeline

The key concept of the given project is the zero knowledge RAG pipeline. Simply put, the system is programmed to respond to questions by using the official academic policy documents. It is not reliant on general model knowledge as it responds. This will assist in ensuring that all responses are grounded on facts rather than speculations.

This begins with locating the appropriate information. The system does not only search the database and extract the most relevant pieces of text depending on keywords, but on meaning as well, when a user asks something. The answer is then generated using these parts that have been selected. This is a clear limitation of the model. It can just utilize what has been recalled and in case the information is inadequate then it must not guess.

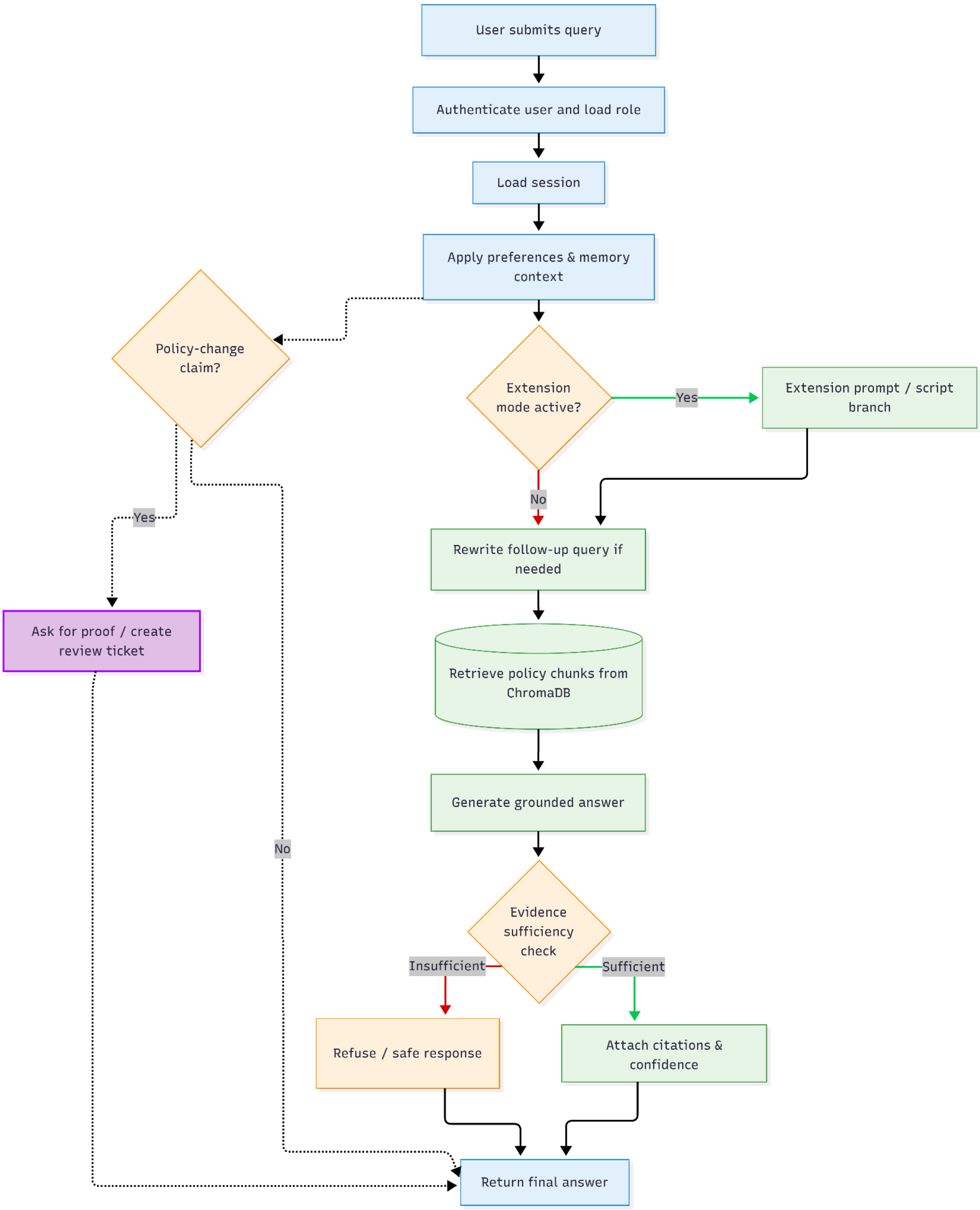

Fig. 5.1: Proposed academic policy assistance system methodological flow.

This strategy is significant in this regard. Academic policies cannot be loosely dealt with. It is not about the soundness, but about being right. And therefore this zero knowledge set-up is not a disadvantage. It is in fact a safety measure which ensures that the system remains on track in the way institutional rules are to be processed.

5.3 Document Ingestion, Chunking, and Retrieval

The system creates its knowledge base by using academic policy documents, which are written in markdown. When they are added they are initially loaded and broken down into smaller bits of text. With this, there are additional items that are stored such as the document it was taken out of, the section name, its status and version information. This not only assists in locating the appropriate content in future, but also in handling other tasks such as updates or unfreezing obsolete sections.

The process or chunking of splitting is done in a careful way. In case the pieces of the text are too large, it will be more difficult to locate the specific useful section. Conversely, when they are too small, not all the crucial context may be lost. The system therefore seeks to maintain a balance hence ensuring every chunk is significant yet accurate enough to be matched.

These chunks are then turned into embeddings and stored in ChromaDB. When a user poses a question, this question is also transformed into an embedding and is compared with the stored ones. Then it selects the most pertinent chunks and works them to come up with the answer. In this manner, the system remains in touch with the real documents and yet, users can pose questions in a natural manner.

5.4 Grounded Response Generation and Refusal Strategy

When the system detects the pertinent policy content, it utilizes the information to produce a response. It is constructed solely out of those parts that are retrieved thus it remains within what is actually present in the documents. This is in order to provide clear and helpful instructions, yet without extending the context provided. In other instances, the system may also indicate the source of the answer and the confidence of the answer which makes life more open.

An important aspect of this process is the ability to know when not to respond. In case the content that has been retrieved is not sufficient to help in coming up with a proper response, the system will just deny it rather than making an educated guess. This may seem harsh, but it will prevent dissemination of wrong or unsubstantiated information.

In a sense therefore, the system is based on two thoughts. It is responsive only when there is substantial evidence and it is not responsive when there is not. This is a balance that is vital particularly in academic environments where a minor error concerning rules or processes can be the cause of confusion.

5.5 Session History and Query Rewriting

In practice, individuals do not necessarily ask the full questions. They tend to pose questions such as What about ESE? or Can you tell me more? Such follow-up questions rely heavily on the previous conversations. So the system will monitor the conversation and make use of that history to know what the user actually wants.

However, it is not always sufficient to look at the previous messages. The question is sometimes too brief or broad and this complicates identifying the correct policy information. In such instances, the system paraphrases the query into a more understandable, fuller form and maintains the original meaning of the query. The revised version is then applied in search of the relevant content and provides the answer.

Due to this fact, the system will be able to cope with back and forth dialogues more easily. Meanwhile, it also abides by the same policy of utilizing retrieved policy content only. The difference is that it operates with a more clear variant of the question of the user, and the results are more precise and helpful.

5.6 Personalization and Secure Memory Framework

A certain degree of personalization is also incorporated into the system so that the responses can be more personal to the individual user. It does not alter the real policy information, but alters the manner in which the information is conveyed. An example is that the system can alter the tone, the format or the amount of detail depending on the manner in which the user typically interacts.

These are preferences which are acquired over time based on the way the users behave. The system keeps them and the degree to which it is certain that there are such patterns. They could mirror concepts such as the manner in which a user would like to be explained or the way the responses are organized. When generating answers, these preferences are applied, but the content still stays fully based on the policy documents.

Memory is also controlled by the system along with this. Information stored is only associated with the particular user and does not spread to other users. Users will also be in control of what to keep, update, or delete. There is, then, no storage of anything without some user control.

Finally, the two aspects of personalization and memory are handled thoughtfully. They exist to make the system more useful but never supersede the actual policy content or violate user privacy.

5.7 Authentication and Identity Verification

The system also comes with authentication thus acting as a real institutional platform, rather than an open chatbot accessible by anyone. Users must authenticate with Google OAuth, with the backend authenticating users with secure tokens. This arrangement assists the system to monitor user sessions, comprehend user roles, and block access to some functionality when necessary.

It also has an additional procedure of user verification, especially non-admin. Here, the users will need to have some institutional evidence. This is checked by an AI-based process before full access is granted through the system. This is a filter that ensures that the user is in fact a member of the institution and is at the appropriate level of access.

In general, the system does not simply concentrate on question answering. It also ensures that users are authenticated and managed appropriately depending on the role. This is very much unlike the normal chat systems, which do not normally validate identity or impose such restrictions.

5.8 Extension-Based Academic Assistance

The platform is designed in a manner that it is capable of more than simply responding to policy questions. It also supports the so-called extensions enabling it to cope with other academic assignments. These are of two kinds. One is prompt-based, whereby custom instructions are provided to be used in particular cases. The other is a script based and in this case the system can execute controlled Python workflows to do more elaborate tasks.

This proves handy since in the academic settings, where a chatbot is not enough, it has to deal with things that cannot be managed by a normal chatbot. Activities such as grading analysis, plotting of CO and PO or even creation of question papers do not lend themselves

To deal with this, the platform includes a structured update process. If a user thinks a policy has changed or something is incorrect, they can raise it. The system then looks at the claim, asks for supporting proof, and creates a review request. After that, authorized to a basic question answer. The system can be extended to include all of these in a single location rather than having to create distinct tools that would be utilized.

Due to this, the platform is made more flexible. It can be easily extended and used in other purposes. Therefore, it is not confined to retrieving policies only but it can be a more comprehensive academic support system.

5.9 Controlled Knowledge Evolution and Policy Update Workflow

The ability of the system to deal with policy changes with time is one of the key characteristics of the system. Rules do not always remain the same in academic institutions. New circulars arrive, procedures are revised and occasionally older policies become obsolete. Unless the system can manage this, its information can easily be invalid.

To address this, the site has a systematic updating process. When a user believes that a policy has been changed or there is something wrong, he can raise it. The system then reviews the claim indicating that it requests supporting evidence and generates a review request. Subsequently, authorized reviewers are involved. They verify the evidence, determine what sections of the stored policy are impacted, and determine whether any of the content should be revised or deleted.

The point here is that, it is all under human control. The system does not automatically alter the policy when someone informs about a problem. Rather, it assists in organizing the process identifying claims, managing evidence and tracking changes, with the final decision remaining with the reviewers.

In this manner, updates are treated with caution. The system is also correct in the long run, and simultaneously, there is appropriate control and documentation of what has been changed and the reasons!

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 Frontend Implementation

React and Vite are used to create the front end of the system and make it a role-based academic interface. It is not to be used anonymously. When an individual logs in, the person will be redirected to a setup, which is the same as the position of the respective individual. This may contain such features as chat, preferences, memory, extensions, policy updates, and the pages related to the administration based on their access.

For most users, the chat section is the main place of interaction. It encourages continuous dialogue, allowing users to continue on past messages. It is also able to support file uploads on demand. The responses shown here are based on policy documents, along with citations and some indication of confidence. When the system does not know, or cannot locate sufficient information, it will give out a warning or not give an answer at all, rather than giving out a questionable response.

The sections of preferences and memory management are also separate. Users have the option to see what the system has come to know about their likes or dislikes, amend the same in case of necessity or delete the stored data. This renders the adaptive features more transparent and the control remains within the grasp of the user.

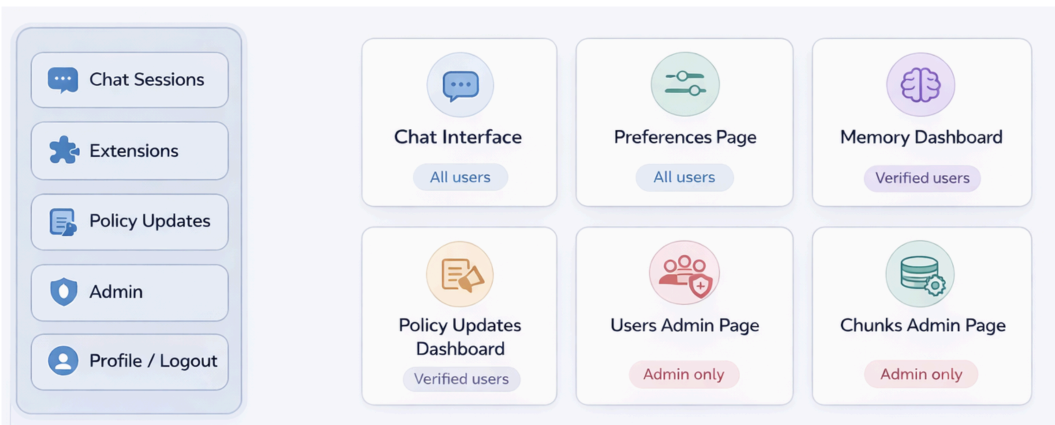

Fig. 6.1: Frontend module layout of the implemented academic policy assistance platform.

There is also the frontend that has extensions and policy management tools. Extension-related features can be accessed by faculty and administrators and policy update requests by reviewers via a dashboard. This dashboard displays the unfinished items, associated proofs, and enables the actions of reviewing it in an organized manner. On the whole, the interface is designed in such a way that it helps in various activities as depicted in Fig. 6.1.

6.2 Backend Implementation

FastAPI is used to create the backend of the system, and it essentially serves as the core of the machine that powers all the things. It offers various API paths to deal with such issues as a login, user access, chat sessions, messages, system health, memory, extensions, and updates of policies. As each of the everything is arranged as a separate module, then various parts such as user handling, retrieval and administrative tasks can be handled in isolation without getting confused.

The main logic is processed in the backend by various services. On receiving a query, it is processed in a pipeline and loads the conversation context and identifies the relevant policy content, applies any user preferences where required and then produces a response based upon that. In parallel with this, there are distinct components to process user, extract and apply preferences, handle memory, execute extensions, detect policy contradictions, search documents, maintain audit records and even update or rollback policy data.

The backend also ensures that access is adequately managed. It verifies authentication with the help of a token, that is, only authorized users are able to access secure sections of the system. Certain paths are further limited according to user roles, particularly the reviewers or administrators. It can also be used to upload files which come in handy when providing evidence when updating a policy or when performing an extension based task.

In general, the backend is designed in such a manner that it ensures everything is organized and safe and various tasks are addressed in the distinct layers as illustrated in Fig. 6.2.

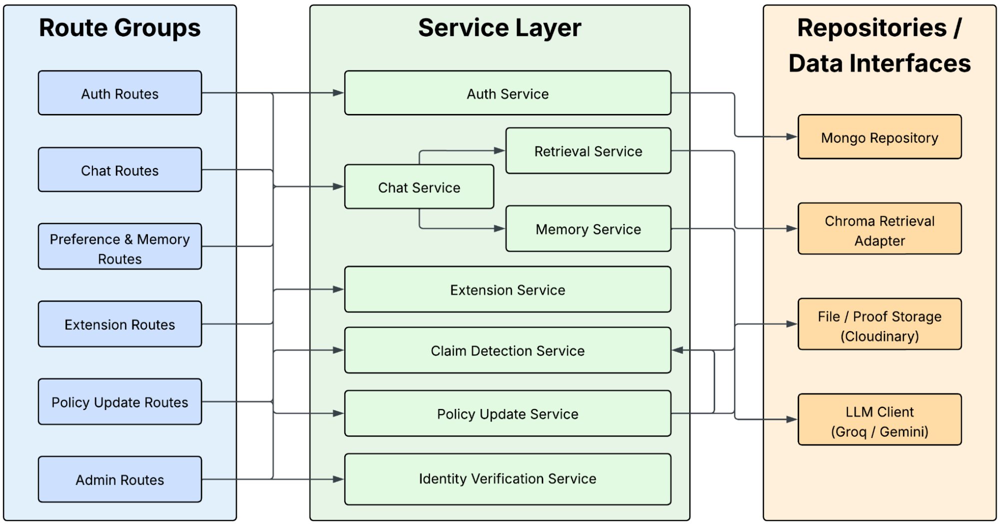

Fig. 6.2: Architecture of backend routes and services in the adopted FastAPI system.

6.3 Database and Storage Implementation

The system employs two storage systems, each having a role. Application-related data is processed by MongoDB, and ChromaDB is utilized to work with policy documents and retrieval. This division aids in maintaining user data and policy knowledge in a well-structured position rather than in a jumble of all.

In MongoDB, things which change with time are the primary things stored. This contains user information, chat history, messages, preferences, memory, extensions and policy update history. As such data may change significantly in form, such a flexible format as MongoDB is suited. To illustrate, a policy update ticket could have varying information at each stage, and the data in memory can vary too, depending on user interaction.

Conversely, the policy content processed is stored in ChromaDB. The documents are fragmented, turned into embeddings, and stored with information about the document they are a part of, its section, and whether it is inactive or outdated. This additional information is helpful when policies are changed, as old pieces can be distinguished and processed without making changes to the rest of the system.

The segregation of these two kinds of data is a significant aspect of the design. It simplifies the management of the system and assists in making sure that user-related operations of the system as well as the retrieval of policy proceeds smoothly.

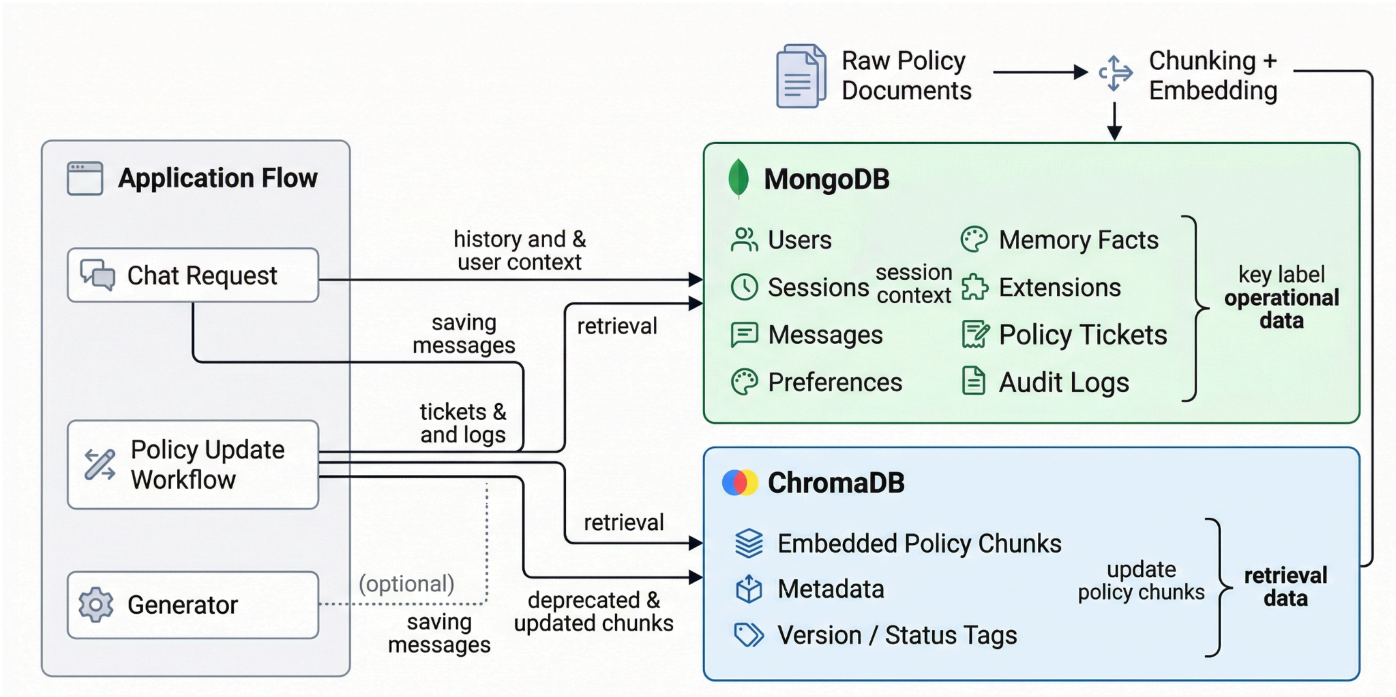

Fig. 6.3: Data interaction flow in the implemented platform between MongoDB and ChromaDB.

The difference between application data and knowledge-base data is the key of the system design. An overview of this part is presented in Table 6.1, and the flow of data interaction in Fig. 6.3.

Storage Component	**Main Purpose**	**Stored Data**
MongoDB	Operational application storage	Users, sessions, messages, preferences, memory facts, extensions, policy tickets, audit data
ChromaDB	Vector-based policy retrieval	Embedded policy chunks, retrieval metadata, status/version information

Table 6.1: Storage role of MongoDB and ChromaDB in the proposed system.

6.4 API Integration and Workflow

The frontend and backend are connected via REST APIs. Simply put, the frontend makes requests of various actions such as asking questions, user profile management, preference, memory, extension, or policy review. These requests are passed to the backend who then processes them and interacts with MongoDB or ChromaDB accordingly and responds with a structured response.

One of the main flows here is the chat process. Frontend will send the question and session details when a user poses a question. The backend then comes in. It verifies the history of the session, locates the corresponding content of the policy and constructs a grounded response. Provided that the user provides a file, it is processed using a multipart request. The identical arrangement is applicable to extensions requiring files uploads or additional execution information.

The same case applies to policy updates which have more steps. The frontend dashboard displays items, such as tickets, proofs and review actions, but the actual work occurs via backend APIs. There are separate endpoints for listing tickets, viewing details, approving or rejecting them, finding affected policy sections, and updating outdated content. In this manner, the frontend remains straightforward, and all the significant logic and control is managed by the backend. Table 6.2 indicates the key application workflows that are being applied using API integration.

Workflow	**Frontend Action**	**Backend Responsibility**	**Result**
Chat query	Send question and optional file	Retrieve chunks, build context, generate grounded response	Answer with citations/confidence or refusal
Preference update	Submit preference change	Store and manage preference state	Personalized response behavior updated
Memory control	Confirm, retain, or forget facts	Update user-specific memory records	Controlled adaptive memory handling
Extension usage	Trigger extension session or upload file	Execute prompt/script extension flow	Domain-specific academic support output

Workflow	Frontend Action	Backend Responsibility	Result
Policy review	Open, inspect, approve, reject, or implement ticket	Manage governance workflow and chunk operations	Controlled policy knowledge evolution

Table 6.2: Key frontend-backend integration processes in the deployed platform.

6.5 Security and Role-Based Controls

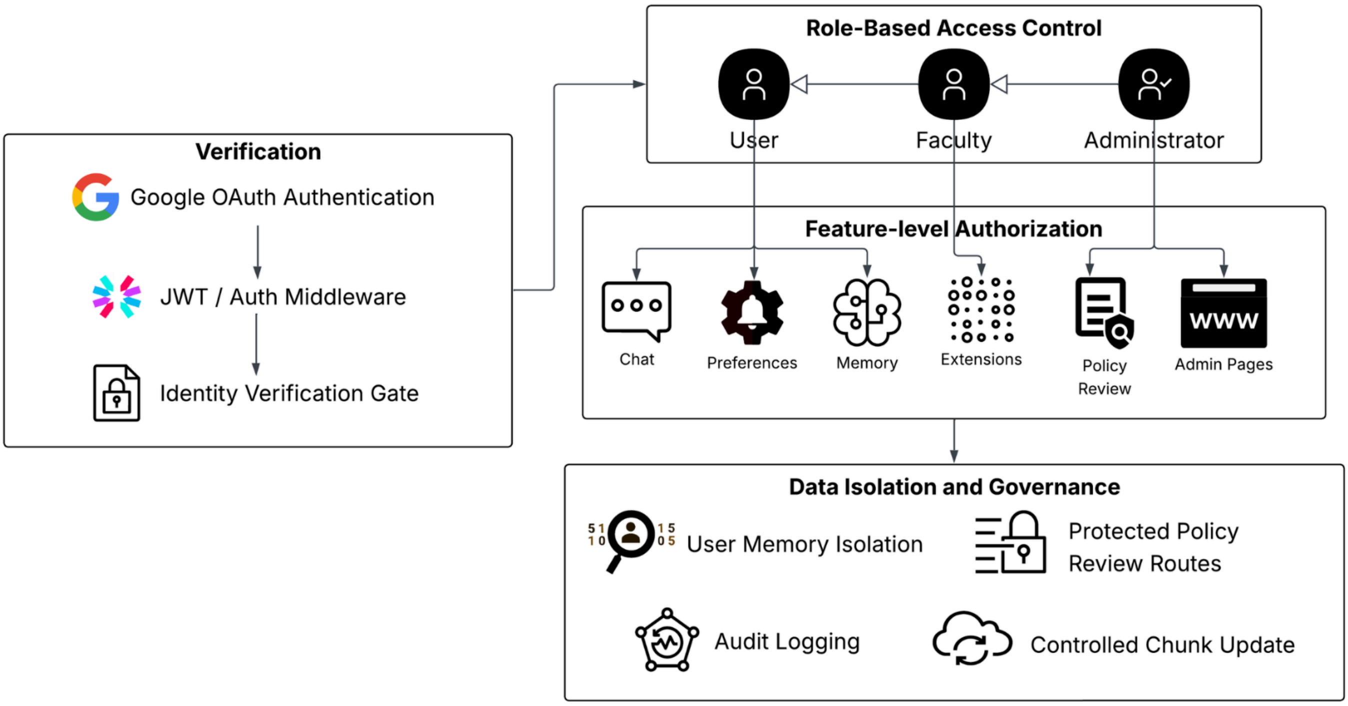

Fig. 6.4: Role-control and layer security model of the implemented system.

The system is multi-layered, rather than being based on a single approach to security. It starts with user authentication. Individuals use Google OAuth to log in, and once this is verified, the backend authenticates the identity of an individual with the help of secure tokens. This will ensure that only the verified users are allowed to access secured sections of the system.

Besides that, it has role-based access control. All users do not receive the same features. The access is varied among students, faculty and administrators. As an example, the chat, preferences and memory functionality can be used by the regular user, whereas the reviewing of policy changes or handling of proofs are restricted to higher-level roles. This will aid in avoiding abuse and makes sensitive actions limited.

The next aspect of significance is the way in which data is processed. Data such as memory and preferences specific to users are stored at the individual level and are not made available to users across the world. Moreover, backend routing of actions such as uploading proof, reading policies, or running extensions do so with validated backend routes. They do not just come about during ordinary conversation.

So overall, the system uses a layered approach. It integrates the process of logging in, role management, and cautious data management to ensure that nothing becomes unsafe and yet users are able to communicate with the system without any complications. The project employs a layered security model as depicted in Fig. 6.4 and the interrelationship between the user roles and access to the features is described in Table 6.3.

User Role	**Accessible Features**
Student	Chat, sessions, preferences, memory controls, identity verification, limited policy claim submission
Faculty	Student features, faculty-oriented extensions, reviewer-level policy workflows where permitted
Administrator	Full platform access including users, chunks, extensions, policy dashboard, and governance actions

Table 6.3: Distribution of role-based access to the key features in the system.

CHAPTER 7

VALIDATION AND EXPERIMENTATION

7.1 Experimental Setup

The verification and testing phase of this project is concerned with the verification of the data and the system itself. Because this is a document-based academic assistant, it does not suffice that the answers have a good sound. The system must also employ the appropriate data, remain rooted in the real policy material and act properly when used in reality.

To make it simple, the validation is split into two. The former section examines the data. In this case, the academic policy documents are verified to ensure that they are well organized, significant, and retrievable. With a weak data, the system may not work well regardless of the quality of the model.

The second section is concerned with the functionality of the application. This involves testing the query processing following a login, the query maintenance, and display of answers grounded, displaying refusals when necessary, and the actual behavior of such features as personalization, memory, and updating policies in the field.

And so this step does not simply test something. It examines the quality of the data, as well as the running of the system. These, collectively, provide a better understanding of the trustworthiness of the overall academic assistant.

7.2 Dataset Description

It begins with a body of knowledge comprising of ten academic policy documents. The papers are specifically written towards the Nova Crest Institute of Engineering and therefore the contents are specific to the institutional setting.These are documents which are canonical corpus of document ingestion, document chunking, document embedding, and document retrieval in the zero knowledge retrieval augmented generation pipeline.

The following academic areas are represented in the dataset:

- academic advice and mentoring,
- continuous assessment overview,
- add, drop, and withdrawal of courses,
- exam registration process,

- final year project guidelines,
- grading elements and weight,
- internship registration and assessment,
- laboratory examination and in-house marks,
- and project submission and review flow, and
- review and revaluation and answer script review.

These papers were authored in a formatted markdown style and were meant to mimic real-life academic policy behavior. Their document structure, sectioning, and procedural content were subsequently checked to be consistent, realistic and meet retrieval augmented generation criteria with an LLM based review procedure incorporated in the project documents.

7.3 LLM-Based Dataset Validation

Claude Sonnet 4.5 was used as an LLM based validator to carry out the dataset validation. Based on the dataset validation report released on January 22, 2026, all ten academic policy documents were assessed on various quality dimensions to answer whether they could be used as a canonical retrieval augmented generation knowledge base. The entire document is contained in the validation materials as the dataset validation report and the key observations are highlighted here to report purposes.

The documents were evaluated by the validator in eight dimensions:

- structural consistency,
- content completeness,
- cross-reference accuracy,
- terminology consistency,
- logical coherence,
- realism and authenticity,
- RAG suitability, and
- data quality.

The overall result of the evaluation was positive. According to the report, the dataset is very consistent and can be implemented as a policy oriented retrieval corpus.The overall validation averages 93.1 percent across the ten documents and the dataset itself is officially designated as being acceptable to use. The report also shows a couple of obvious strengths such as well-structured organization of documents, the natural academic policy voice, and the good preparedness to the semantic retrieval. In general, the level of data quality is quite high. These scores are broken down in detail in Table 7.1.

Validation Dimension	Average Score
Structural Consistency	95.6%
Content Completeness	91.5%
Cross-Reference Accuracy	87.8%
Terminology Consistency	92.1%
Logical Coherence	92.9%
Realism and Authenticity	90.5%
RAG Suitability	95.3%
Data Quality	97.1%
Overall Average Score	**93.1%**

Table 7.1: Overview of LLM-based dataset validation scores with NCIE academic policy corpus (source: Claude Sonnet 4.5 validation report).

Meanwhile, there are some minor problems noted in the validation. Certain documents might be enhanced by clarity in cross-referencing, and some minor differences in the use of certain terms. Nevertheless, these are regarded as small and those that are not critical enough to impact usability.

In the perspective of the project, the dataset is solid, then. It is sound and semantically clear, and thus can be trusted to be used in a zero knowledge academic assistance system.

7.4 Functional Testing

In this project, functional testing is primarily concerned with determining whether every component of the system is really functioning as intended. This involves simple aspects such as receiving user requests, establishing sessions, accessing policy information and providing responses that are well based on the documents. It also verifies that the systemis no supporting evidence, and whether different users see the right features based on their roles.

denies the answer in case there is no supporting evidence and different users view the correct features depending on their positions.

The validation was at the backend level, as opposed to merely testing individual screens or interfaces. The conceptualization was to test the system under real-life situations, with authenticated users and active sessions, and not under isolated tests.

On the application level, a number of areas were provided. This involves access and logging in, creating and maintaining a history of a session, creating answers to acceptable academic policy questions, and putting the unacceptable ones away appropriately. It also verifies the functionality of preferences and memory, availability of extensions depending on role, and functionality of policy update dashboard and ticket review process.

To do testing, a session based benchmark was employed with an integrated validation setup. The benchmark consisted of 22 sessions and 150 turns of conversation. Overall, 153 attempts were made, 151 of which were successfully completed, and 2 were interrupted because of the external provider constraints. The systems did not cause these interruptions and therefore do not influence the judgement of the correctness of the system. The results are elaborated and are reported in both the application core validation report and the scenario flow validation report.

Overall, the results show that the system works as expected from a functional point of view. It manages authenticated access, session flow, generation of grounded answers, refusal where necessary and policy governance properties. It has not only been implemented, but also tested and demonstrated to be useful in actual use situations.

7.5 Retrieval and Response Evaluation

Evaluating retrieval and response quality is important to see if the system really works like an academic policy assistant, instead of just acting like a normal chatbot. The validation of the dataset is the quality of retrieval in this project. The report reveals high ratings in such areas as the structure, logical flow, and readiness to retrieve augmented generation.

The policy documents are well structured with distinct sections and content that can be easily digest into chunks. Each section is dedicated to a particular topic, and it becomes easier to align the user queries with the appropriate policy sections, which is facilitated by the system. Practically, this eliminates confusion and assists the system to locate pertinent answers more precisely.

To test this appropriately, the test was done at the query level. This was done by looking at whether the right documents were retrieved to the common policy questions, whether the best results were really relevant, whether the final results were similar to the evidence retrieved and whether the system failed to answer when no appropriate policy was located.

The overall results from the application validation show that the system performs reliably in these areas. The main measures are:

- grounded response fidelity: 0.6784
- context sufficiency rate: 0.6755
- hallucination rate: 0.0463
- refusal correctness rate: 1.0

These figures indicate that the system largely remains in line with the content of the retrieved policy. The rate of hallucination is very low with respect to the size of the test and the rate of correctly refusing is 1.0 that is the system never gave an answer when it did not have the right evidence. This, in combination, implies that the system is in a controlled and reliable manner. It is concerned with providing evidence-based answers and not coming up with unsubstantiated ones.

7.6 Personalization and Memory Testing

The system does not simply concern the retrieval of correct answers. It also involves the element of personalization and user memory hence, the assessment must do more than merely testing accuracy. It examines whether user preferences are used as intended, whether stored memory remains unique to individual users and whether users are actually able to access and/or delete their data at their will.

In this validation, a number of things were tested. This involves ensuring preferences are captured and stored in the right way and whether responses made using these preferences remain rooted in policy documents. It also examines the ability of the users to access, maintain or delete the memory stored. The other check that is important is that the information of one user remains entirely independent of another and that role-based limitations remain applicable even in the situation of personalization.

The most important outcomes of the overall testing of the application are:

- personalization accuracy: 0.7391
- user memory relevance score: 1.2693
- zero-knowledge compliance: 1.0

- knowledge evolution stability: 1.0

These findings indicate that the system can be personalized to users without necessarily going out of bounds. Especially important is the score of 1.0 on zero-knowledge compliance, as it establishes that personalization does not cause unsupported or assumed information. Equally, the knowledge stability score of 1.0 implies that the system manages the memory and updates reliably as the system is used repeatedly.

7.7 Policy Update Workflow Testing

The policy update feature is a major component of this project thus it had to be tested appropriately on its own. The workflow includes a number of steps, such as identifying the moment when a user reports a potential change or contradiction in the policy, enabling proof to be uploaded, creating a review ticket, discovering which sections of the policy are impacted, and ultimately letting reviewers act.

The entire process was part of the application-level validation as it is one of the key distinguishing characteristics of this system. The assessment considered various points along the workflow. It verified that the system is able to identify a claim by a user regarding a policy issue correctly, that proof files are uploaded and attached correctly, and that tickets are generated and displayed in the dashboard. It also experimented with things such as reviewer actions such as approval and rejection, as well as how the system detects the affected chunk and either updates or removes the outdated content.

It is also a critical part since a majority of RAG-based systems presuppose that the data remains constant. In this case, the system is proactive in dealing with updates. The validation results show:

- contradiction resolution rate: 1.0
- policy claim false alarm rate: 0.0

These results are quite strong. The resolution rate of 1.0 indicates that all valid contradictions cases were treated properly and the false alarm of 0.0 indicates that the system did not generate unneeded updates. So in general, not only is this feature present, but it performs in a predictable and regulated manner during testing.

7.8 Key Results and Observations

Looking at the entire results of the validation, the most important conclusion is that both the data and the system as such are robust enough to prove what this project proclaims.

The policy documents were tested and scored a total of 93.1 percent, and in application testing, the system was able to cope with 151 of 153 attempts in realistic multi-turn conditions.

The application side notes some of the notable observations. The zero-knowledge compliance of the system was 1.0 and this indicates that the system remained disciplined and did not produce unsupported answers. It also scored high on contradiction resolution with a score of 1.0 and stability of knowledge evolution with a score of 1.0 which indicates that the update and governance mechanism is reliable.

The system had a personalization of 0.7391 meaning that the user adaptation is actually in operation not a mere feature. Meanwhile, the rate of hallucination was low (0.0463), and the refusal correctness was 1.0, indicating that the system never gave an answer when there was no appropriate evidence. The other significant observation is that the assistant acted as a system that was aware of a session. It may deal with on-going discussions rather than each query as an individual, isolated query. These findings are further enhanced with the scenario-based benchmark. It consisted of 22 realistic tasks involving things such as memory tests, follow-up questions, preference learning, contradiction management and more complex logic problems. It is due to this that the assessment is much more realistic in regard to conversational application as opposed to mere one-time prompts. Table 7.2 summarizes these results.

Validation Measure	**Result**
Dataset overall validation score	93.1%
Total application attempts	153
Successful application responses	151
Grounded response fidelity	0.6784
Hallucination rate	0.0463
Personalization accuracy	0.7391
Zero-knowledge compliance	1.0
Refusal correctness rate	1.0

Table 7.2: Final summary of data set and application validation results of the final NCIE academic assistant system.

CHAPTER 8

RESULTS AND OUTCOME

8.1 System Outcomes

The ultimate outcome of the project is the development of Zero Knowledge Adaptive AI Agent that can assist in academic policies. It is more than a normal retrieval based chatbot. Rather, it is a coordinated system, which integrates document based answering, secure user access, personalization, extensions, and a controlled manner to update policies.

In its essence, the system demonstrates the feasibility of the zero knowledge policy answering in practice. When a user queries on academic procedures, the system searches the content of relevant policies, and constructs its answer using the documents. This is because when it cannot get sufficient information to support it, it just refuses to answer rather than guess. This is to keep the behavior consistent and applicable in institutions.

Basic question answering is not the only thing offered by the project. It has capabilities such as user authentication and role access, identity verification with AI support, and handling of preferences. It also securely manages memory among multiple users. To top it all, it accommodates extensions to added functionality and a due process of reviewing and updating policies.

All these characteristics demonstrate that it is not a mere chatbot. It is a superior academic support resource that is established to manage policy based interactions in a regulated and programmed manner.

8.2 Accuracy and Reliability Outcomes

The credibility of this project is largely due to two factors: grounded retrieval and quality of source data. The system is built on a retrieval augmented generation strategy, meaning that it produces answers by referring to the real policy documents rather than being backed by the general model memory. Owing to this, the responses remain attached to actual institutional contents, and hence reduce misplaced or unsubstantiated responses.

This reliability is also supported by the validation of the dataset. The policy data have an average of 93.1 percent in LLM based assessment. It fared exceptionally well in such factors as structure, data quality, and its applicability in retrieval-based applications. This implies that the knowledge base of this type of zero knowledge system is organized well and it functions with a zero knowledge system.

Testing the application itself, also shows evidence. The findings indicate a grounded response score of 0.6784 and context sufficiency rate of 0.6755. The hallucination rate is low with the rate being 0.0463 and the refusal correctness rate will be 1.0. These figures demonstrate that the system is not only based on design concepts, but in fact acts in a natural and predictable manner when it is interacting in real-life scenarios.

The other significant aspect of reliability is the way the system deals with cases in which it is unable to respond. It is able to say no when they do not have sufficient information as opposed to making an attempt to answer everything. This proves to be handy as it is usually better to provide no response than provide a wrong response particularly in academic policy cases.

In general, the project demonstrates that one can create an academic assistant, which is both helpful and attentive. It can provide useful answers and be grounded, traceable and safe in its approach to information.

8.3 Practical Significance in Academic Institutions

This project is practical in the real sense as it relates well with real needs in the academic institutions. Colleges and universities frequently have numerous recurrent questions concerning policies, particularly, exams, registration, grading, revaluation, projects, and internships. In most cases, the students and the staff members are required to read documents or make calls to the administration in order to seek answers. This system provides easier and quicker alternative.

It simplifies things among students. They do not have to look in notices or handbooks, but can simply pose a question in everyday language and receive a response founded on official policy documents. The system can also be useful to faculty and administrators. It facilitates the use of roles, enables the use of other tools via extensions, and provides an organized process to overview and maintain policy changes.

The other important aspect is that the system is able to be flexible and remain controlled. It has more user friendly features such as memory and preferences, but they are limited and under the control of the user. Updating of policies is also not carried out freely. They are taken through an adequate review and tracking, which assists in sustaining credibility and accuracy.

Thus, the worth of this project is not necessarily in the mechanism of its functioning. It also demonstrates how institutions can practically utilize AI systems without the risk of losing the authority over correctness, transparency, and general governance.

8.4 Limitations of the Current System

The project achieves its primary objectives, yet it has certain obvious shortcomings that should be mentioned. Firstly, the quality and coverage of academic policy data is important to the extent to which the system will perform. Unless certain policies are found in the documents, are unclear or not updated, the system might fail to provide an appropriate response. In some sense therefore the output is as good as the data it is constructed on.

The other one is that the system is made to be controlled in a more academic use and demonstration, rather than large scale production. Certain sections are made to be accurate and well structured, as opposed to fast or extensive use. In addition, the update of the policy must be reviewed by a human. This not only prevents the workflow but also makes the work safe and reliable, as well as introduces a little bit of added work.

There are also a few ideas that were not included in the current system. Such issues as anticipating the risk in academics or integrating vast amount of knowledge were taken into consideration but they do not form part of what has been introduced in this case. These should rather be perceived as potential future directions rather than completed features.

Despite these restrictions, the project demonstrates an effective attitude towards supporting academic policies. It is able to remain safe, based on real data, and well regulated in its information management and update protocol.

CHAPTER 9

CONCLUSION AND FUTURE WORK

9.1 Conclusion

This project focused on building and implementing a Zero Knowledge Adaptive AI Agent for helping with academic policies. The system was made out to be a controlled support platform whereby a number of components such as retrieval augmented generation, secure multi-user access, personalization, identity checks, and extension support are incorporated into one configuration in addition to the structured manner of updating policies.

The idea started from a common issue. Information on academic policies is typically found in documents and administrative routes and is difficult to locate. To deal with this, the project used a zero knowledge retrieval approach. This made sure the system depends only on official policy documents instead of general model memory. Due to this, the answers remain anchored to real-life sources and even the system would not give the answers where there is a lack of evidence.

In addition to the primary retrieval system, several additional layers were included. Authentication and role-based access assisted in making the system more than a simple chatbot. It turned out to be an official administrative instrument. These characteristics such as user preferences and secure memory enabled the system to adapt to its responses and still providing control to the users. Verification of identity provided an added level of trust, particularly to non-admin users. The extension environment demonstrated that the platform is capable of managing a greater number of academic assignments, and the policy update procedure ensured that all updates are assessed and monitored rather than implemented arbitrarily.

The policy knowledge base was also tested with an LLM based evaluation using the dataset used to create the policy knowledge base. It was found that it can be used in a retrieval-based system. Making it even worse, the very application was tested using session testing on the back end. These tests revealed that the system is capable of dealing with real and multi-step interactions with appropriate authentication, grounded responses, memory manipulation, and controlled updates. The fact that the dataset, the system and the various scenarios of using the system are independently validated gives the work even greater strength.

Ultimately, the project demonstrates that academic policy support does not necessarily need to be based on a general chatbot. It may be constructed as a system that is secure, which is based on real data and which is properly maintained. The ultimate output is more of an institutional knowledge site with restricted flexibility and human management, as opposed to a mere chatty tool.

9.2 Future Enhancements

The project is achieving what is supposed to be achieved but there are still some areas which can be enhanced with time. The first step that will be obvious is the extension of the institutional knowledge base. It currently is effective with the data available, but it would be more comprehensive with the inclusion of additional policy documents, circulars, past revisions and department-specific rules. This would also enable the system to provide answers to a broader scope of questions without going through gaps.

Memory and personalization is another area. It is currently effective, but it can be adjusted to be more adaptable. The next generation would allow users to manage their data more effectively and the system would be able to deal with trust in information stored in a more efficient manner. It would also be beneficial to make users able to easily see how the system evolves over time. It is also possible to better the policy update process. As an example, the detection of conflicts, the management of supporting evidence, and helping reviewers could all be enhanced. This would be easier to manage with better matching methods, more robust audit tools and straightforward dashboards to monitor changes.

The extension system has room to grow as well. It already supports both prompt based and script based extensions so can be expanded into a more extensive tool set. The system could be extended with more faculty and administrative features that could be used within the same system.

Moving forward, some interesting directions can be searched. The system may be shifted to predictive academic support or even enable what-if like policy reasoning. More advanced features for institutional use are also possible. Nevertheless, adding them, it is necessary to preserve the overall concepts of the project. The system must be kept down to earth, easy to operate, user controlled and prudent in updating its knowledge.

Bibliography

1. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. *Advances in Neural Information Processing Systems*, 33. https://arxiv.org/abs/2005.11401

2. Izacard, G., & Grave, E. (2021). *Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering*. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (pp. 874–880). Association for Computational Linguistics. https://aclanthology.org/2021.acl-long.479/

3. Graves, A., Wayne, G., & Danihelka, I. (2014). *Neural Turing Machines*. https://arxiv.org/abs/1410.5401

4. Parisi, G. I., Kemker, R., Part, J. L., Kanan, C., & Wermter, S. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks*, 113, 54–71. https://ieeexplore.ieee.org/document/8673865

5. Bourtoule, L., Chandrasekaran, V., Choquette-Choo, C. A., Jia, H., Travers, A., Zhang, B., Lie, D., & Papernot, N. (2021). *Machine Unlearning*. In *2021 IEEE Symposium on Security and Privacy (SP)* (pp. 141–159). IEEE. https://ieeexplore.ieee.org/document/9519395

6. Zhang, S., Dinan, E., Urbanek, J., Szlam, A., Kiela, D., & Weston, J. (2018). *Personalizing Dialogue Agents: I Have a Name, I Like to Be Called*. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics. https://aclanthology.org/P18-1205/

7. Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021). *Retrieval Augmentation Reduces Hallucination in Conversation*. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 3784–3803). Association for Computational Linguistics. https://aclanthology.org/2021.emnlp-main.357/

8. Dwork, C., & Roth, A. (2014). *The Algorithmic Foundations of Differential Privacy*. *Foundations and Trends in Theoretical Computer Science*, 9(3-4), 211–407. https://www.nowpublishers.com/article/Details/TCS-042

9. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 610–623). ACM. https://dl.acm.org/doi/10.1145/3442188.3445922

10. Ghazvininejad, M., Brockett, C., Chang, M. W., Dolan, B., Gao, J., Yih, W. T., & Galley, M. (2018). *A Knowledge-Grounded Neural Conversation Model*. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1). https://arxiv.org/abs/1702.01932

11. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, 30. https://papers.nips.cc/paper/7181-attention-is-all-you-need

12. Hardt, D. (Ed.). (2012). *The OAuth 2.0 Authorization Framework* (RFC 6749). Internet Engineering Task Force. https://datatracker.ietf.org/doc/html/rfc6749

13. Grassi, P., Newton, E., Perlner, R., Regenscheid, A., Burr, W., Richer, J., Lefkovitz, N., Danker, J., Choong, Y. Y., Greene, K., & Theofanos, M. (2017, updated 2020). *Digital Identity Guidelines: Authentication and Lifecycle Management* (NIST SP 800-63B). National Institute of Standards and Technology. https://csrc.nist.gov/pubs/sp/800/63/b/upd2/final

14. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., & Lowe, R. (2022). *Training language models to follow instructions with human feedback*. https://arxiv.org/abs/2203.02155

15. Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N. S., Chen, A. S., Creel, K., Davis, J. Q., Demszky, D., et al. (2021). *On the Opportunities and Risks of Foundation Models*. Stanford Center for Research on Foundation Models. https://crfm.stanford.edu/report.html

16. Yu, H., Gan, A., Zhang, K., Tong, S., Liu, Q., & Liu, Z. (2024). *Evaluation of Retrieval-Augmented Generation: A Survey*. *CoRR*, abs/2405.07437. https://doi.org/10.48550/arXiv.2405.07437

Appendix

Acronyms and abbreviations used

Acronym	Full Form	Description
AI	Artificial Intelligence	Computational systems designed to perform tasks that typically require human intelligence.
API	Application Programming Interface	Interface that enables communication between software modules.
DB	Database	Structured storage system for application data.
HTML	HyperText Markup Language	Standard markup language used for web page structure.
HTTP	HyperText Transfer Protocol	Protocol used for communication between web clients and servers.
ID	Identity Document / Identifier	Used in the project for user verification and system identification.
JWT	JSON Web Token	Token format used for authenticated API access and authorization.
LLM	Large Language Model	Language model used for generation, validation, and workflow assistance.
NCIE	Nova Crest Institute of Engineering	Academic institution context used for the policy corpus in this project.

Acronym	Full Form	Description
NLP	Natural Language Processing	AI field related to language understanding and generation.
OAuth	Open Authorization	Standard authorization protocol used for Google-based login.
PDF	Portable Document Format	Document format used for report files, policy proofs, and submission material.
QA	Question Answering	Task of answering natural language questions.
RAG	Retrieval-Augmented Generation	Technique combining document retrieval with answer generation.
RBAC	Role-Based Access Control	Access-control model based on user roles such as student, faculty, and administrator.
UI	User Interface	Visual interface through which users interact with the system.
URL	Uniform Resource Locator	Web address used for navigation or linking resources.

Appendix A: Validation Materials Included

The following validation materials have been prepared as part of the final report set:

- dataset validation report,
- application core validation report,
- scenario-flow application validation report

Appendix A.1: Dataset Validation Report

Dataset Validation Report: NCIE Academic Policy Documents

Validator: Claude Sonnet 4.5 (LLM)
Validation Date: January 22, 2026
Dataset Source: LLM-generated (ChatGPT)
Total Documents: 10
Purpose: Phase 1 RAG system knowledge-base validation

Executive Summary

All 10 academic policy documents were validated across 8 quality dimensions. The dataset demonstrates high consistency, strong structural discipline, and clear suitability for retrieval-augmented generation. Minor improvements are recommended for cross-referencing and terminology standardization, but no critical issues were identified.

Overall Rating: Approved for production use

Validation Methodology

Each document was evaluated across the following dimensions:

1. **Structural Consistency**: format, sections, and organization
2. **Content Completeness**: coverage of topic scope
3. **Cross-Reference Accuracy**: internal document references
4. **Terminology Consistency**: standard terms across documents
5. **Logical Coherence**: policy logic and flow
6. **Realism and Authenticity**: resemblance to real academic policies
7. **RAG Suitability**: chunking, retrieval, and clarity
8. **Data Quality**: grammar, formatting, and accuracy

Page 1

Detailed Validation Results

#	DOCUMENT NAME	STRUCTURAL	COMPLETENESS	CROSS-REF	TERMINOLOGY	COHERENCE	REALISM	RAG SUITABILITY	QUALITY	OVERALL SCORE
1	academic_advising_and_mentorship.md	95%	90%	85%	95%	95%	90%	95%	98%	**93%**
2	continuous_assessment_overview.md	98%	95%	90%	95%	95%	95%	98%	98%	**96%**
3	course_add_drop_and_withdrawal.md	95%	92%	88%	92%	92%	90%	95%	96%	**93%**
4	exam_registration_process.md	98%	95%	92%	95%	95%	92%	98%	98%	**95%**
5	final_year_project_guidelines.md	95%	90%	85%	90%	92%	88%	95%	96%	**91%**
6	grading_components_and_weightage.md	98%	95%	95%	98%	98%	95%	98%	98%	**97%**
7	internship_registration_and_evaluation.md	95%	88%	82%	90%	90%	85%	92%	96%	**90%**
8	lab_evaluation_and_internal_marks.md	95%	92%	88%	92%	95%	92%	95%	98%	**93%**
9	project_submission_and_review_flow.md	92%	88%	85%	88%	90%	88%	92%	95%	**90%**
10	revaluation_and_answer_script_review.md	95%	90%	88%	92%	92%	90%	95%	98%	**93%**

Average Overall Score: 93.1%

Dimension-Wise Analysis

1. Structural Consistency (95.6% avg)

Rating: Excellent

Strengths:

- All documents follow a consistent markdown structure with numbered sections.
- The section hierarchy is clear and easy to navigate.
- Section numbering is uniform and sequential.
- Heading styles are consistent across the dataset.

Observations:

- Nine of the ten documents use the heading "Introduction"; one uses "Overview".
- All documents include a closing summary section.
- Average document depth is appropriate for policy-style material.

Issues: None critical

2. Content Completeness (91.5% avg)

Rating: Excellent

Strengths:

- All documents cover their intended scope well.
- The dataset includes workflow, roles, systems, and exception-handling perspectives.
- The level of detail is suitable for RAG-based question answering.

Topic Coverage Matrix:

DOCUMENT	WORKFLOW	ROLES	EXCEPTIONS	SYSTEMS	TIMELINE
Academic Advising	Yes	Yes	Yes	Yes	Partial
Continuous Assessment	Yes	Yes	Yes	Yes	Yes
Course Add/Drop	Yes	Yes	Yes	Yes	Partial
Exam Registration	Yes	Yes	Yes	Yes	Yes
Final Year Project	Yes	Yes	Yes	Yes	Yes
Grading Components	Yes	Yes	Partial	Yes	N/A
Internship Evaluation	Yes	Yes	Yes	Yes	Yes
Lab Evaluation	Yes	Yes	Yes	Yes	Partial
Project Submission	Yes	Yes	Yes	Yes	Yes
Revaluation Process	Yes	Yes	Yes	Yes	Partial

Minor Gaps:

- Specific calendar dates are intentionally abstracted.
- Some documents do not include concrete examples, which is acceptable for policy-style writing.

3. Cross-Reference Accuracy (87.8% avg)

Rating: Good

Strengths:

4. Terminology Consistency (92.1% avg)

Rating: Excellent

Standard Terms Used Consistently:

TERM	FREQUENCY	CONSISTENCY	NOTES
Nova Crest Institute of Engineering (NCIE)	10/10	100%	Always bold and expanded on first use
Continuous Assessment (CA)	8/10	100%	Consistent abbreviation
End-Semester Examination (ESE)	6/10	100%	Consistent abbreviation
Faculty Advisor	8/10	100%	Consistent role reference
Department Academic Coordinator	9/10	100%	Consistent role reference
Examination Cell	4/10	100%	Consistent usage
Project Guide / Supervisor	2/2	100%	Interchangeable but acceptable
Academic portal	10/10	100%	Lowercase and consistent
Academic system	10/10	100%	Lowercase and consistent

Minor Inconsistencies:

- Capitalization varies slightly in some mid-sentence references.
- "Project Guide" and "Project Supervisor" are both used in one project-related document.

Recommendation: Standardize capitalization conventions for mid-sentence terminology.

5. Logical Coherence (92.9% avg)

Rating: Excellent

Strengths:

- Cause-and-effect relationships are clear.

- Workflows follow a logical sequence.
- No contradictions were found across the dataset.
- Process descriptions are internally coherent.

Illustrative Process Flow:

```
Student -> Faculty Advisor -> Department Coordinator -> System
Update Portal -> Eligibility Check -> Approval -> Record Update
```

Coherence Checks:

POLICY LOGIC	DOCUMENT	STATUS
CA (40%) + ESE (60%) = 100%	grading_components	Correct
Dropped courses are not treated as completed academic record items	course_add_drop	Correct
Withdrawn courses are not exam-eligible in the normal way	course_add_drop	Correct
Revaluation applies to ESE rather than CA	revaluation	Correct
CA is tied to enrollment and course activity	continuous_assessment	Correct
Lab marks contribute within the CA framework	lab_evaluation	Correct
Final year project spans Semesters 7 and 8	final_year_project	Correct
Internships are evaluated differently from normal course grading	internship	Correct

Issues: None found

6. Realism and Authenticity (90.5% avg)

Rating: Excellent

Strengths:

- The policies resemble realistic engineering-college academic documentation.
- The tone is appropriately institutional.
- Roles and workflows are credible.

- The procedural structure matches common academic-administration patterns.

Realism Comparison with Real Academic Institutions:

POLICY ELEMENT	NCIE DATASET	REAL INSTITUTIONS	ASSESSMENT
CA:ESE ratio	40:60	Common range 30:70 to 50:50	Realistic
Course add/drop process	Portal-based	Portal or manual	Realistic
Revaluation mechanism	ESE-focused	Common practice	Accurate
Faculty Advisor role	Guidance-focused	Common practice	Accurate
Final-year project span	Semesters 7 and 8	Standard in many Indian institutions	Accurate
Examination Cell model	Centralized	Common practice	Accurate
Internship as program component	Yes	Common practice	Realistic

Minor Authenticity Gaps:

- No detailed grade-range table is included.
- Fee amounts are intentionally omitted.
- Institutional status details are not specified.

Verdict: The dataset is highly authentic for an academic-policy knowledge base.

7. RAG Suitability (95.3% avg)

Rating: Excellent

Strengths:

- Sections are well structured for chunking.
- Topic sentences are clear and retrieval-friendly.
- Sections are generally self-contained.
- Titles and headers are explicit and semantically useful.

Chunking Analysis:

DOCUMENT	AVG SECTION LENGTH	CHUNK QUALITY	SELF-CONTAINED
academic_advising	110 words	Good	Yes
continuous_assessment	105 words	Good	Yes
course_add_drop	95 words	Good	Yes
exam_registration	120 words	Good	Yes
final_year_project	100 words	Good	Yes
grading_components	90 words	Excellent	Yes
internship	110 words	Good	Yes
lab_evaluation	95 words	Good	Yes
project_submission	95 words	Good	Yes
revaluation	100 words	Good	Yes

Optimal Chunk Size: 90 to 120 words per section

Semantic Search Testing:

QUERY	EXPECTED DOCUMENT	RETRIEVAL LIKELIHOOD
"How do I register for exams?"	exam_registration	High
"What is the CA weightage?"	grading_components	High
"Can I drop a course?"	course_add_drop	High
"Faculty advisor role"	academic_advising	High
"Final year project timeline"	final_year_project	High
"Revaluation process steps"	revaluation	High

RAG Compatibility: Excellent. The documents are RAG-ready without structural modification.

8. Data Quality (97.1% avg)

Rating: Excellent

Grammar and Language:

- No spelling issues of significance were identified.
- The professional academic tone is maintained consistently.
- Verb tense is stable and appropriate.
- Sentences are clear and unambiguous.

Formatting Quality:

ASPECT	STATUS	NOTES
Markdown syntax	Strong	Headings and lists are well formed
Bullet formatting	Consistent	List style is stable
Bold formatting	Correct	Key terms are emphasized appropriately
Section numbering	Sequential	No numbering issues detected
Line breaks	Proper	Spacing is consistent

Metadata Quality:

- Titles are clear and descriptive.
- No duplicate document content of concern was found.
- No placeholder text remained.
- No broken references were observed.

Cross-Document Consistency Matrix

Key Metrics Consistency Check

METRIC	DOCUMENT 1	DOCUMENT 2	DOCUMENT 3	CONSISTENT?
CA Weightage	40% (grading)	40% (continuous_assessment)	40% (lab_evaluation)	Yes
ESE Weightage	60% (grading)	60% (continuous_assessment)	60% (exam_registration)	Yes
Project Duration	Sem 7-8 (final_year)	Sem 7-8 (project_submission)	Sem 7-8 (internship)	Yes

METRIC	DOCUMENT 1	DOCUMENT 2	DOCUMENT 3	CONSISTENT?
Faculty Advisor Role	Guidance (advising)	Approval (course_add_drop)	Guidance (exam_registration)	Yes
Revaluation Scope	ESE only (revaluation)	ESE only (grading)	CA not included (continuous_assessment)	Yes

Role Definitions Consistency

ROLE	DOCUMENT	DEFINITION	CONFLICT?
Faculty Advisor	academic_advising	Guides course selection and monitors progress	No
Faculty Advisor	course_add_drop	Reviews add/drop requests	No
Faculty Advisor	exam_registration	Clarifies eligibility	No
Dept Coordinator	continuous_assessment	Oversees consistency	No
Dept Coordinator	course_add_drop	Oversees approvals	No
Examination Cell	exam_registration	Manages registration	No
Examination Cell	revaluation	Coordinates revaluation	No

Result: No role-definition conflicts were detected.

Critical Issues and Recommendations

Critical Issues

None detected. The dataset is suitable for production use.

Recommended Improvements

1. **Cross-reference enhancement**
    - Some documents could link more explicitly to related policies.
    - Example: the revaluation document could directly reference exam registration.
    - Impact: Low
2. **Terminology capitalization**
    - Some mid-sentence capitalization is slightly inconsistent.
    - Impact: Very low
3. **Concrete examples**
    - Optional examples or FAQ-style notes could improve interpretability.
    - Impact: Low
4. **Timeline specificity**
    - Some timeline phrases remain intentionally abstract.
    - Impact: Low

Strengths To Maintain

1. Consistent institutional voice and tone
2. Clear structure that is well suited to RAG chunking
3. Self-contained sections with minimal cross-dependency
4. Appropriate abstraction level for policy documentation
5. Professional academic language throughout

RAG System Integration Assessment

Embedding Quality Prediction

DOCUMENT	SEMANTIC CLARITY	KEYWORD DENSITY	EMBEDDING SCORE
academic_advising	High	Medium	92/100
continuous_assessment	Very High	High	98/100

DOCUMENT	SEMANTIC CLARITY	KEYWORD DENSITY	EMBEDDING SCORE
course_add_drop	High	High	95/100
exam_registration	Very High	High	98/100
final_year_project	High	Medium	90/100
grading_components	Very High	Very High	99/100
internship	Medium-High	Medium	88/100
lab_evaluation	High	High	95/100
project_submission	High	Medium	90/100
revaluation	High	High	95/100

Average Embedding Score: 94/100

Retrieval Performance Prediction

QUERY TYPE	EXAMPLE QUERY	EXPECTED RETRIEVAL	CONFIDENCE
Direct Match	"What is continuous assessment?"	continuous_assessment.md	98%
Process Query	"How to register for exams"	exam_registration.md	95%
Numeric Query	"CA weightage percentage"	grading_components.md	97%
Role Query	"Faculty advisor responsibilities"	academic_advising.md	93%
Comparison	"Difference between drop and withdrawal"	course_add_drop.md	92%
Complex	"Can I get my exam answer sheet reviewed?"	revaluation.md	90%

Predicted Top-K Accuracy:

- Top-1: 94%
- Top-3: 98%
- Top-5: 99%

Chunk Overlap Analysis

CONCEPT	DOCUMENT 1	DOCUMENT 2	OVERLAP %	ISSUE?
CA Definition	continuous_assessment	grading_components	15%	No
ESE Definition	exam_registration	grading_components	12%	No
Faculty Advisor	academic_advising	course_add_drop	8%	No
System Integration	Multiple documents	N/A	5-10%	No

Result: No problematic content duplication was identified.

Validation Conclusion

Overall Assessment

Dataset Quality: Excellent
Structural Quality: 95.6%
Content Coverage: 91.5%
Consistency: 92.1%
Authenticity: 90.5%
RAG Readiness: 95.3%
Data Quality: 97.1%

Overall Score: 93.1% (A grade)

Validation Verdict

The LLM-generated dataset demonstrates:

1. High structural and content quality
2. Excellent consistency across documents
3. Realistic representation of academic policies
4. Strong structure for RAG implementation
5. Minimal cross-document conflicts

Recommended Actions

Before Deployment:

1. No critical fixes are required.
2. Optional: add explicit cross-reference links.
3. Optional: standardize capitalization conventions.

For Future Iterations:

1. Consider adding FAQ sections for common queries.
2. Add concrete examples or case studies if desired.
3. Include specific timeline dates if an institutional calendar becomes available.

Comparative Analysis: LLM-Generated vs Human-Written

ASPECT	LLM-GENERATED (THIS DATASET)	TYPICAL HUMAN-WRITTEN DOCS
Consistency	Excellent	Variable
Completeness	Excellent	Good
Formatting	Excellent	Variable
Terminology	Highly consistent	Moderately consistent
Realism	Very good	Authentic
Specific Examples	Limited	Rich
Edge Cases	Good	Comprehensive
RAG Suitability	Optimal	Variable

Conclusion: The LLM-generated documents show superior structural consistency and strong RAG suitability compared with typical human-written policy documents, with minor trade-offs in authenticity detail and concrete examples.

Validation Signatures

Validator: Claude Sonnet 4.5 (Anthropic)
Validation Method: Multi-dimensional automated analysis
Documents Analyzed: 10/10
Total Validation Time: Approximately 15 minutes
Validation Date: January 22, 2026

Final Recommendation: Approved for Phase 1 RAG system deployment

This validation report indicates that LLM-generated academic policy documents, when created carefully and validated systematically, can meet a high standard for use in RAG-based academic assistant systems.

Appendix A.2: Scenario-flow Application Validation Report

Scenario Flow Validation Report: GPT-Authored Multi-Turn Benchmark

Scenario Authoring: Codex / GPT-5.4-assisted benchmark design
Validation Date: April 1, 2026
Benchmark Type: Multi-session natural-conversation application validation
Benchmark Size: 22 sessions, 150 authored turns
Purpose: Validate the assistant under realistic conversational flows rather than isolated policy questions

Executive Summary

To evaluate the assistant in a realistic way, the benchmark was intentionally designed as a sequence of healthy, natural conversations rather than as disconnected single-turn prompts. The sessions begin with simple onboarding and profile learning, then progress through preference learning, grounded academic-policy questions, memory checks, clarification pressure, contradiction handling, policy-proof workflows, and finally dense reasoning-heavy scenarios.

This scenario design is one of the strongest aspects of the project because it validates the assistant as a conversational system with continuity. Instead of asking only "Can the model answer this question?", the benchmark asks "Can the application maintain context, use memory appropriately, stay grounded, and still behave well over time?"

Overall Rating: Strong scenario-based benchmark suitable for report and presentation use

Why A Scenario-Based Benchmark Was Necessary

A normal QA benchmark would not capture the central behavior of this application. The NCIE assistant is designed to:

- learn user preferences gradually
- remember user facts across turns
- interpret follow-up questions through session context

- refuse unsupported answers
- surface contradictions and governance workflows when needed

Because of that, the benchmark itself had to be conversational. This is why the questions were grouped into sessions rather than shuffled into a flat list.

Benchmark Structure

COMPONENT	VALUE
Total sessions	22
Total authored turns	150
Session style	Multi-turn, natural conversation
User profile pattern	Reused and evolved across sessions
Evaluation mode	Backend execution with live validation

The benchmark begins with normal student-style interactions and gradually increases complexity.

Early sessions

These sessions focus on:

- greetings and fresh-session zero-knowledge behavior
- natural self-introduction
- preference learning
- profile recall
- basic academic workflow questions

Middle sessions

These sessions test:

- consistent personalization across follow-ups
- grounded answering under reformulated questions
- user confusion and correction flows

- policy-change and proof-upload logic
- contradiction-aware reasoning

Final sessions

The last benchmark block is intentionally reasoning-heavy. These sessions combine multiple constraints at once, such as:

- project plus internship scheduling
- attendance plus assessment implications
- grading plus revaluation logic
- memory plus personalized advising boundaries
- policy claims plus evidence quality
- out-of-scope requests mixed with grounded academic guidance

This makes the benchmark closer to real use than a standard FAQ-style evaluation.

Session Design Quality

The scenario benchmark was designed to feel like genuine student conversation. It avoids robotic prompt templates and instead uses:

- natural introductions
- style preferences stated conversationally
- short follow-up turns such as "Are you sure?" and "Then what about ESE?"
- explicit memory checks such as asking whether the assistant knows the user
- realistic confusion around grading, revaluation, proof, and policy updates

This matters because conversational continuity is exactly where many systems appear good in demos but break in practice.

Coverage Areas

The authored scenario set covers the main behavioral requirements of the system:

COVERAGE AREA	INCLUDED IN BENCHMARK
Groundedness	Yes
Hallucination resistance	Yes
Completeness of response	Yes
Context carryover	Yes
Preference learning	Yes
Memory use	Yes
Zero-knowledge behavior	Yes
Contradiction handling	Yes
Policy-claim workflow	Yes
Refusal correctness	Yes
Dense reasoning	Yes

This breadth is important because it shows that the benchmark was not built only to make the system look good. It was built to expose how the system behaves across the full range of intended features.

Benchmark Progression From Start To End

1. Session entry and user setup

The benchmark starts with ordinary conversation, including greeting, user introduction, and natural expression of response-style preference. This establishes whether the assistant can begin safely and then become personalized through interaction rather than through hidden setup.

2. Stable academic-policy conversation

The next block focuses on standard NCIE topics such as:

- grading components

- project guidelines
- internship registration
- exam registration
- add/drop and withdrawal
- lab evaluation and internal marks

These sessions verify the assistant's ability to answer grounded academic questions in an ongoing session.

3. Memory and behavior checks

Once the user profile and preference are learned, the benchmark probes whether the system:

- remembers facts naturally
- avoids overclaiming memory
- uses the right tone later in the session
- remains context-aware when the user asks abbreviated follow-ups

4. Governance and contradiction workflows

The scenario design then moves beyond simple policy QA into application logic:

- what happens if a user says a policy changed
- when proof should be requested
- how conflicting facts are handled
- how academic-policy claims should be escalated

This is the part that makes the benchmark application-oriented rather than purely document-oriented.

5. Reasoning-heavy capstone sessions

The final session block combines multiple academic factors in one turn. These are not simple retrieval prompts. They require the assistant to:

- understand multi-factor student situations
- connect relevant parts of retrieved policy context
- remain grounded while still being useful
- preserve session continuity under longer conversations

This provides a realistic upper-bound test of the assistant's intended behavior.

Why This Benchmark Is Strong For A Final-Year Project

This scenario benchmark strengthens the project in three important ways.

1. It validates the application, not only the model

The benchmark exercises the live backend, session store, memory behavior, retrieval, generation, and validator together. That makes it much more valuable than a prompt-only test set.

2. It demonstrates system design maturity

Grouping 150 turns into 22 realistic sessions shows that evaluation was treated as part of the architecture. That is a strong systems-engineering decision and is academically valuable in its own right.

3. It supports meaningful reporting

Because the benchmark is session-based, the report can discuss:

- how the assistant starts
- how it learns
- how it remembers
- how it clarifies
- how it reasons

- how it behaves under policy uncertainty

That creates a much better narrative than a flat list of isolated question-answer pairs.

Interpretation Of The Scenario Validation

The scenario benchmark shows that the assistant can sustain realistic multi-turn academic conversations while preserving the project's main design goals:

- document grounding
- adaptive personalization
- controlled memory usage
- conversational continuity
- policy-governance awareness

A very small number of interrupted attempts occurred during long-run execution because of provider usage limits, but this does not materially change the benchmark's value or structure. The scenario set itself remains valid, coherent, and representative of intended use.

Final Assessment

The GPT-authored multi-turn benchmark is a strong validation asset for this project. It moves the evaluation beyond single-answer correctness and into session-based behavioral validation, which is the right way to assess an assistant that includes memory, preference learning, and policy-governance features.

In summary, the scenario-flow validation shows that the project was evaluated:

- conversationally rather than superficially
- behaviorally rather than only textually
- across the full application lifecycle rather than isolated turns

Final Verdict: Validated as a strong scenario-driven benchmark for natural-flow application evaluation

Validation Signatures

Scenario Authoring: Codex / GPT-5.4-assisted benchmark design
Validation Method: Session-based multi-turn benchmark evaluation over live backend execution
Benchmark Scope: 22 sessions and 150 authored turns
Validation Date: April 1, 2026

Final Recommendation: Approved as a strong scenario-flow validation report for natural conversational evaluation

This validation report indicates that the NCIE academic assistant was evaluated through a realistic multi-turn benchmark designed to reflect natural student conversation, allowing context continuity, memory behavior, personalization, and reasoning quality to be assessed in a structured and repeatable way.

Appendix A.3: Application Validation Report

Application Core Validation Report: NCIE Academic Assistant

Validator: OpenAI `gpt-4.1-mini`
Validation Date: April 1, 2026
System Under Test: Authenticated NCIE academic-policy assistant with RAG, memory, preference learning, contradiction handling, and policy-claim workflow
Validation Source: Integrated application validation summary
Purpose: Core application-response validation across the end-to-end backend workflow

Executive Summary

The application was validated as a full backend system rather than as an isolated answer generator. The evaluation covered authenticated querying, session continuity, retrieval-grounded answering, user-memory behavior, preference application, contradiction handling, and policy-governance logic. Across the integrated validation set, the system completed **151 successful responses out of 153 total attempts**, with the remaining two interruptions attributable to provider usage limits rather than application logic defects.

Overall, the application demonstrates a strong and credible response core for an adaptive academic assistant. It is especially strong in zero-knowledge discipline, contradiction handling, knowledge evolution stability, refusal correctness, and low false-alarm behavior for policy claims. The response core is therefore suitable to present as a working, evaluated system rather than a prototype with only anecdotal testing.

Overall Rating: Approved for application-level evaluation and report presentation

Validation Scope

The core validation covered the real application flow from start to finish:

1. User authentication with a live JWT-authenticated account
2. Session creation through the backend API
3. User query submission through the production query route

4. Retrieval from the academic policy knowledge base
5. Response generation with personalization and memory hooks
6. Per-turn validation using a separate LLM judge
7. Rolling aggregation into report artifacts

This means the report reflects the actual system behavior under realistic execution, not hand-curated answer examples.

Validation Methodology

The application was evaluated with a two-layer validation approach.

1. Deterministic checks

Each turn was checked for:

- session continuity
- source presence
- latency capture
- refusal alignment where applicable
- preference extraction/application signals
- fact-memory changes
- policy-claim detection behavior
- retry and recovery behavior

2. LLM-judge evaluation

A separate validator model reviewed the query, retrieved sources, response, and behavioral metadata to score:

- groundedness
- hallucination behavior
- relevance
- completeness
- context sufficiency

- memory relevance
- personalization correctness
- contradiction handling quality
- overall accept/reject verdict

3. Integrated reporting

The final application summary used the integrated files from:

- `test/reports/till-avs17/application_validation_full`
- `test/reports/after_17/application_validation_full`

These were merged into a single weighted summary in:

- `test/reports/integrated/application_validation_full`

Validation Volume

MEASURE	VALUE
Total attempts	153
Successful responses	151
Interrupted attempts	2
Effective success rate	98.69%
Average latency	43271.82 ms
Conservative p95 latency	74405.15 ms

The two interrupted attempts were associated with external provider usage limits during generation and do not materially affect the interpretation of the application's behavioral quality.

Core Metrics

METRIC	VALUE	INTERPRETATION
Personalization Accuracy	0.7391	The assistant usually adapted its style and behavior correctly once user preferences were learned.

METRIC	VALUE	INTERPRETATION
Zero-Knowledge Compliance	1.0	The assistant stayed disciplined when no prior knowledge should be assumed.
User Memory Relevance Score	1.2693	Learned facts were used meaningfully, though not always maximally.
Grounded Response Fidelity	0.6784	Most accepted answers remained tied to retrieved material, with room to tighten explicit support.
Contradiction Resolution Rate	1.0	Contradictory policy reasoning was handled reliably in evaluated cases.
Knowledge Evolution Stability	1.0	Learned and updated knowledge remained stable across relevant turns.
End-to-End Task Success Rate	0.6536	The system completed a strong majority of benchmark goals in full conversational context.
Hallucination Rate	0.0463	Hallucinations were relatively infrequent in the integrated run.
Context Sufficiency Rate	0.6755	Retrieved context was often adequate, though some responses could still be more explicit or better synthesized.
Retry Recovery Rate	0.3333	The system was able to recover from part of the transient execution pressure it encountered.
Refusal Correctness Rate	1.0	Out-of-scope or unsupported responses were refused appropriately in the relevant cases.
Policy Claim False Alarm Rate	0.0	No spurious policy-claim escalation was introduced in the validated flow.

What The Core Validation Shows

1. The application behaves as a real adaptive assistant

This is not a single-turn QA bot. The validated backend can:

- learn a user's conversational preference
- preserve session context across turns
- recall user details when relevant
- keep answers grounded in academic-policy documents

- identify when claims require proof-based governance

That makes the application materially stronger than a plain retrieval demo.

2. The system is strongest in safety and governance discipline

The most reassuring outcomes in the core validation are:

- perfect zero-knowledge compliance
- perfect contradiction resolution in tested cases
- perfect knowledge-evolution stability in tested cases
- perfect refusal correctness in measured refusal scenarios
- zero policy-claim false alarms

These outcomes support the claim that the system is not only informative, but also behaviorally controlled.

3. The system's main opportunity is answer sharpness, not conceptual correctness

The integrated metrics suggest that the remaining gaps are mostly in:

- making grounded answers more explicit
- improving completeness on follow-up clarification turns
- using memory more fully when the user explicitly asks for recall

This is a refinement problem, not a foundational architecture problem.

Operational Notes

The validation also exercised realistic execution pressure, including provider-side usage limits. The application and harness were updated so that provider-limit events can be surfaced in machine-readable form without changing the user-facing fallback behavior. This allowed the validation process to remain faithful to the real workflow while still producing trustworthy evaluation data.

As a result, the core report should be interpreted as a backend validation of the actual system behavior under realistic constraints, not as an idealized offline benchmark.

Final Assessment

The NCIE academic assistant demonstrates a credible, well-structured response core with strong behavioral control. It performs especially well in safety-sensitive areas such as zero-knowledge behavior, contradiction handling, refusal correctness, and governance stability. Personalization and memory are meaningfully present, and the integrated success rate is strong enough to support the system as a serious final-year project artifact.

In summary, the core application validation supports the claim that the system is:

- grounded rather than purely generative
- adaptive rather than stateless
- controlled rather than loosely conversational
- evaluable through repeatable backend-driven validation

Final Verdict: Validated as a strong application-response core for the NCIE academic assistant

Validation Signatures

Validator: OpenAI `gpt-4.1-mini`
Validation Method: Backend-driven application validation with deterministic checks and LLM-judge review
Validation Scope: 153 total attempts across integrated application validation runs
Successful Responses: 151
Validation Date: April 1, 2026

Final Recommendation: Approved as a strong core application validation report for the NCIE academic assistant

This validation report indicates that the NCIE academic assistant has been evaluated as a full application system, with grounded-response behavior, personalization, memory usage, contradiction handling, and governance-aware workflow all assessed through repeatable backend validation.

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

2% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

22 Not Cited or Quoted 2%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

1%	Internet sources
1%	Publications
2%	Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **22** Not Cited or Quoted 2%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

1% Internet sources
1% Publications
2% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1 Student papers
PSG Institutions on 2025-11-06 <1%

2 Student papers
PSG Institutions on 2025-05-09 <1%

3 Internet
www.coursehero.com <1%

4 Internet
eprints.utar.edu.my <1%

5 Student papers
Informatics Education Limited on 2008-11-30 <1%

6 Student papers
University of Hertfordshire on 2026-03-26 <1%

7 Student papers
Kingston University on 2025-09-18 <1%

8 Student papers
Majan College on 2014-06-02 <1%

9 Student papers
Informatics Education Limited on 2009-03-19 <1%

10 Student papers
Radboud Universiteit on 2025-07-16 <1%

11 Student papers

University of Birmingham on 2025-04-17 <1%

12 Student papers

University of Hertfordshire on 2026-03-19 <1%

13 Student papers

University of Ulster on 2012-05-04 <1%

14 Student papers

University of Westminster on 2025-05-05 <1%

15 Student papers

University of Wollongong on 2024-12-04 <1%

16 Internet

huggingface.co <1%

17 Internet

ijecs.in <1%

18 Student papers

Liverpool John Moores University on 2025-05-15 <1%